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MILITARY SPECIFICATION

TEST-EQUIPMENT, FOR USE WITH ELECTRONIC EQUIPMENT: GENERAL SPECIFICATION

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments

1. SCOPE

1.1 This specification covers the general requirements for the design and manufacture of test equipment used in testing of electronic equipment. The detail requirements for a particular test set are covered in the individual specification for that test set.

2. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS, AND DRAWINGS

2.1 **Specifications.**—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

FEDERAL SPECIFICATIONS

QQ-M-151—Metals; General Specification for Inspection of.
QQ-S-571—Solder; Soft (Tin, Tin-Lead, and Lead-Silver).
CCC-T-191—Textiles; General Specifications, Test Methods.

MILITARY SPECIFICATIONS

JAN-1—Tubes, Electron.
JAN-C-5—Capacitors, Mica-Dielectric, Fixed.
JAN-I-6—Instruments, Electrical Measuring, Indicating, Basic, Voltmeters and Ammeters.
JAN-I-7—Insulators, Glass-Bonded-Mica, Radio.
JAN-I-8—Insulators, Steatite, Radio.
JAN-I-9—Insulators, Glass, Radio.
JAN-I-10—Insulating Materials, Ceramic Radio, Class L.
JAN-R-11—Resistors, Fixed, Composition.
JAN-I-12—Insulating Materials, Ceramic Radio Dielectric, Class H.
JAN-P-13—Plastic-Materials, Laminated, Thermosetting: Sheets and Plates.
JAN-P-14—Plastic-Materials, Molded, Thermosetting.
JAN-C-17—Cables, Coaxial and Twin-Conductor, for Radio Frequency.
JAN-B-18—Batteries, Dry.
JAN-R-19—Resistors, Variable Wire-Wound (Low Operating Temperature).
JAN-C-20—Capacitors, Fixed, Ceramic-Dielectric (Temperature-Compensating).
JAN-I-21—Insulators, Porcelain, Radio.
JAN-R-22—Rheostats, Wire-Wound, Power-Type.
JAN-S-23—Switches, Toggle (for Electronic and Communications Use).
JAN-C-25—Capacitors, Direct-Current, Paper-Dielectric, Fixed (Hermetically Sealed in Metallic Cases).
JAN-R-26—Resistors, fixed, Wire-Wound Power-Type.
MIL-T-27—Transformers and Inductors (Audio and Power).
JAN-S-28—Sockets, Electron Tube and Accessories.
JAN-R-29—Resistors, External Meter (High Voltage, Ferrule Terminal Type).
JAN-S-57—Switches, Vacuum.
JAN-C-62—Capacitors, Dry-Electrolytic, Polarized.
JAN-S-63—Switches, Sensitive.
JAN-C-76—Cable (Hook-Up Wire), Electric, Insulated, Radio and Instrument.

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JAN-P-77—Plastic-Materials, Cast, Thermosetting.
JAN-P-78—Plastic-Materials, Laminated, Thermosetting (for Name, Graphic, Chart, Instruction, and Designating Plates).
JAN-P-79—Plastic-Materials, Laminated, Thermosetting: Rods and Tubes.
JAN-P-80—Plastic-Materials, Anti-Electrostatic, Clear, Transparent (for Indicating-Instrument Windows).
JAN-C-81 — Capacitors, Ceramic-Dielectric, Variable.
JAN-T-85—Tubing, Waveguide, Seamless Rectangular.
JAN-C-91—Capacitors, Paper-Dielectric, Fixed (Nonmetallic Cases).
JAN-C-92 — Capacitors, Air-Dielectric, Variable (Trimmer Capacitors).
JAN-R-93—Resistors, Accurate, Fixed, Wire-Wound.
JAN-R-94—Resistors, Variable, Composition.
JAN-T-152—Treatment, Moisture- and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment: General Process for.
JAN-C-173 — Coating-Materials, Moisture- and Fungus-Resistant, for the Treatment of Communications, Electronic and Associated Electrical Equipment.
JAN-R-184 — Resistors, Fixed, Wire-Wound (Low-Power).
JAN-W-287 — Wave-Guide-Assemblies, Flexible.
JAN-J-641—Jacks, Telephone.
JAN-P-642—Plugs, Telephone.
JAN-A-669 — Anti-Seize Compound, White Lead Base, General Purpose (For Threaded Fittings).
MIL-C-915—Cables, Electric (Shipboard Use).
MIL-P-997—Plastic-Material, Laminated, Thermosetting, Electrical-Insulating: Sheets, Type GSG.
MIL-C-5015—Connectors; Electrical.

AIR FORCE-NAVY AERONAUTICAL SPECIFICATIONS

AN-C-128—Compound; Insulating and Sealing, Electrical Connections.
AN-F-14—Felt; Wool.
AN-F-15—Felt; Hair.
AN-M-12—Magnesium Alloy; Process for Corrosion Protection of.
AN-N-11—Nut; Instrument Mounting.
AN-N-12—Nut; Sheet Spring.
AN-P-32—Plating; Zinc.
AN-P-34—Plating; Nickel.
AN-P-39—Plating; Chromium.
AN-P-51—Petrolatum.
AN-P-61—Plating; Cadmium.
AN-S-24—Screws; Self-Tapping Steel.
AN-S-61—Shafts; Tuning, MC-215.
AN-T-85—Test Points; Requirements for (for Conventional Pulse Type, X-Band Radar).
AN-QQ-A-696 — Anodic-Films; Corrosion-Protective (for) Aluminum Alloys.

U. S. ARMY SPECIFICATIONS

8-15—Felt, Pressed.
71-3310—Wires WS-17/U, WS-18/U and WS-( )/U.
72-53—Finishes (for Ground Signal Equipment).
100-46—Mildew Resistance and Moisture Resistance Treatments for Felt, Wool.

U. S. ARMY SIGNAL CORPS TENTATIVE SPECIFICATIONS

71-3064—Variable Composition Resistors.
72-78—Plastic-Materials, Rigid, Thermoplastic.

NAVY DEPARTMENT SPECIFICATIONS

General Specifications for Inspection of Material.[1]
16C38—Coils, Choke, Radio-Frequency (for Electronic Use).
16T36—Transformers, Intermediate and Radio Frequency (for Electronic Use).
17P8—Plastic-Materials, Rigid, Thermoplastic.
17C13—Connectors and Lugs, Terminal, Pressure-Grip (Solderless) (for Electric Cable).
17E11—Enclosures (for Nonrotating Electrical Equipment) (Shipboard Use).
17P25—Plastic-Material, Laminated, Thermosetting, Electrical-Insulating: Sheets, Types GMG and GSG.

[1] Applicable only to Navy purchases.

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52E14—Enamel, Equipment, Light-Gray (Formula 111).

(*Army.*—Copies of specifications should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(*Navy.*—Copies of Federal, Military (except JAN-1), and Navy Department specifications may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that activities of the Armed Forces should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. Copies of Specification JAN-1 may be obtained upon application to the Armed Services Electro Standards Agency, Fort Monmouth, N. J. Copies of Air Force Navy Aeronautical (AN) specifications may be obtained upon application to the Commander, U. S. Naval Air Development Center, Johnsville, Pa. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(*Air Force.*—Copies of Federal, Military (except JAN-1), and Air Force-Navy Aeronautical specifications may be obtained upon application to the Commanding General, Air Matériel Command, Wright-Patterson Air Force Base, Dayton, Ohio. Copies of Specification JAN-1 may be obtained upon application to the Armed Services Electro-Standards Agency, Fort Monmouth, N. J. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 Other publications. The following publications, of the issue in effect on date of invitation for bids, form a part of this specification:

MILITARY STANDARD

JAN-STD-15—Electrical and Electronic Symbols.

JOINT ARMY-NAVY PUBLICATIONS

JANP 109—Manual of Standard Descriptions for Electronic Equipment and Material.

Preferred List of Army-Navy Electron Tubes.

Index of Army-Navy R-F Transmission Lines and Fittings.

(*Army.*—Copies of Military standards and Joint Army-Navy publications should be obtained from the procuring agency or as directed by that agency Both the title and identifying number or symbol should be stipulated when requesting copies.)

(*Navy.*—Copies of Military standards and Joint Army-Navy Publication JANP 109 may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that activities of the Armed Forces should make application to the Supply Officer in Command, Naval Supply, Center, Norfolk 11, Va. Both the title and identifying number or symbol should be stipulated when requesting copies. Copies of the Preferred List of Army-Navy Electron Tubes and the Index of Army-Navy R-F Transmission Lines and Fittings may be obtained upon application to the Armed Services Electro Standards Agency, Fort Monmouth, N. J.)

(*Air Force.*—Copies of Military standards and Joint Army-Navy Publication JANP 109 may be obtained upon application to the Commanding General, Air Matériel Command, Wright-Patterson Air Force Base, Dayton, Ohio. Both the title and identifying number or symbol should be stipulated when requesting copies. Copies of the Preferred List of Army-Navy Electron Tubes and the Index of Army-Navy R-F Transmission Lines and Fittings may be obtained upon application to the Armed Services Electro Standards Agency, Fort Monmouth, N. J.)

AMERICAN WAR STANDARD

C75.13—Dynamotors.

(Copies of American War standards may be obtained upon application to the American Standards Association Incorporated, 70 East Forty-fifth Street, New York 17, N. Y. Both the title and identifying number should be stipulated when requesting copies.)

NATIONAL BUREAU OF STANDARDS PUBLICATION

Handbook H28—Screw-Thread Standards for Federal Services.

(Copies of Handbook H28 may be obtained upon application, accompanied by money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

2.3 Drawings. The following drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

AIR FORCE-NAVY AERONAUTICAL DRAWINGS

AN504—Screw—Round Head Self Tapping Machine.

AN506—Screw—82° Flat Head Self Tapping Machine.

AN530—Sheet Metal Screw—Round Head.

AN531—Sheet Metal Screw—Flat Head.

AN659—Terminal—Electrical Solderless.

AN3195—Fastener—Snapslide.

AN3196—Stud—Style I Snapslide Fastener.

AN3197—Stud—Style II Snapslide Fastener.

AN3198—Stud—Style III Snapslide Fastener.

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AN3199—Stud—Style IV Snapslide Fastener.

AND10082—Fastener Installation—Snapslide.

U. S. Army Signal Corps Drawings

SC-C-6878—Test Set—CX-1331/U.

SC-D-6883—Cord—CX-1332/U.

SC-D-15800—Oscilloscope BC-000, Circuit Label (Schematic Diagram).

SC-D-15801—Junction Box JB-000, Circuit Label (Wiring Diagram).

SC-D-15914—Humidity Test Cycle for Ground Signal Equipment.

SC-D-20095—Universal Telephone Cord Circuit, Sample Circuit Label, Schematic Diagram.

SC-D-20096—Rectifier PP-0/0 Sample Circuit Label Schematic Circuit.

SC-D-20097—Radio Set AN/000, Sample Cording Diagram.

Bureau of Ships Drawing

10-T-2145—Light Weight Shock Testing Machine.

Bureau of Ships Radio Drawings

RE49F246—Matched Impedance Connectors, Bayonet Locking, Plug UG-88/U, Jack UG-89/U, Panel Jack UG-291/U, for Use with RG-58/U Cable.

RE49F402—Improved Type N Plugs and Jacks.

(*Army.*—Copies of U. S. Army Signal Corps Drawings may be obtained upon application to the Commanding Officer, Philadelphia Signal Corps Procurement District (SPSAD-X), 128 N. Broad Street, Philadelphia 2, Pa. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(*Navy.*—Copies of Air Force-Navy Aeronautical drawings may be obtained upon application to the Commanding Officer, U. S. Naval Air Development Station Johnsville, Pa. Copies of Bureau of Ships drawings and Bureau of Ships Radio drawings may be obtained upon application to the Bureau of Ships, Navy Department, Washington 25, D. C. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(*Air Force.*—Copies of Air Force-Navy Aeronautical drawings may be obtained upon application to the Commanding General, Air Matériel Command, Wright-Patterson Air Force Base, Dayton, Ohio. Both the title and identifying number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *General.*—The materials, parts, and mechanical assemblies used in the construction of test equipment, but not specified in detail, shall be of a quality consistent with the proposed use and specified performance of the test set. The largest possible proportion of the material used shall be noninflammable. Where there is any doubt as to the proper use of any material, part, or mechanical assembly not specified in detail, such matters shall be referred to the bureau or agency concerned for decision. The employment of design which will tend to minimize the size and weight of the complete test set is a major consideration, and shall be used to the greatest practicable extent.

3.1.2 *Corrosion resistance.*—Materials shall be of corrosion-resisting types or shall be suitably processed to resist corrosion.

3.1.2.1 Gold, nickel, chromium, rhodium, corrosion-resisting steel (12 percent or more chromium), tin, lead-tin alloys, or sufficiently thick platings of these metals are satisfactory without additional protection or treatment other than buffing or cleaning. Aluminum, magnesium, brass, iron, steel, except as specified, cadmium, and zinc, in general, require additional protection (see 3.35).

3.1.3 *Dissimilar metals.*—Dissimilar metals having an electrolytic potential difference of greater than 0.4 volt between them when immersed in 3 percent sodium chloride solution shall not be used with their untreated surfaces in immediate contact with each other. This potential difference (0.4 or less) may in many cases be obtained by electroplating with a suitable metal. The potential difference between the plating metal and the base metal may be greater than 0.4 volt provided the electroplating is continuous over all surfaces of the base metal. Before parts made of the following commonly used metals are assembled in contact with each other, they shall be electroplated with a suitable metal to obtain this potential or painted in accordance with 3.35:

a. Copper alloys against aluminum alloys or zinc base alloys.

b. Corrosion-resisting ferrous alloys against aluminum alloys or zinc base alloys.

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c. Magnesium alloys against aluminum alloys, copper alloys, or platings of silver, cadmium, nickel, or zinc. (See 3.17.)

3.1.4 *Toxic materials.*—The use of materials which are known to produce harmful toxic effects under the conditions normally encountered in military service will not be permitted unless a suitable nontoxic compound is not available. In particular, the use of such materials as rubber hydrochloride which liberates chlorine, shall be avoided.

3.1.5 *Fungus-inert materials.* — Materials which are not nutrients for fungus shall be used to the greatest extent practicable.

3.1.5.1 In general, the following types or kinds of materials shall not be used unless no other materials possessing greater fungus inertness and satisfactory for the purpose intended are available:

Cotton
Linen
Cellulose nitrate
Regenerated cellulose
Wood
Leather
Paper and cardboard
Cork
Hair or wool felts
Plastic materials using cotton, linen or wood flour as a filler.

3.1.5.1.1 If it is necessary to use any of the materials specified in 3.1.5.1, they shall be treated in accordance with the applicable paragraphs of this specification or their use shall be approved by the bureau or agency concerned.

3.1.5.1.2 *Felt.*—Where the use of felt is necessary, it shall be in accordance with Specification 8–15, AN–F–14, or AN–F–15, and shall be suitable for the purpose intended.

3.1.5.1.3 *Wood.*—Where necessary, wood suitably treated for protection against moisture and fungus may be employed for transformer, field and armature wedges, and in other applications where permitted or specified in the individual specification for the particular test set.

3.1.5.2 The following types or kinds of materials shall be used in preference to those specified in 3.1.5.1:

Metals
Ceramics (steatite, glass, glass-bonded-mica, etc.)
Mica
Plastics using glass, mica or asbestos as a filler
Cellulose acetate
Nylon
Polyvinylchloride
Rubber (natural or synthetic)

3.1.6 Zinc parts shall not be used where the increased electrical resistance of the surface due to the protective treatment specified in 3.35 has a deleterious effect on electrical performance.

3.1.7 *Substitution of parts and materials.*—If a contractor desires to substitute another part or material where the specification and drawings call for a specific item, or where a particular part or material has been approved previously for use in the test set, he shall first submit to the bureau or agency concerned a statement to that effect, describing the proposed substitution and reason therefor, and shall submit evidence that such a substitution in the particular test set being furnished is at least the equal of the specific part or material specified or previously approved. At the discretion of the bureau or agency concerned, samples may be required which will demonstrate by test the suitability of the proposed substitution. Parts for which substitutions may be submitted are as follows:

a. A specific part or material.

b. A specific part or material, or "equal", or "equivalent".

c. A specific product of a particular manufacturer or "equal".

3.2 Definitions.—For the purpose of this specification, the following definitions shall apply:

3.2.1 *Transit case.*—A transit case is a case without a test set built in, having a compartment for accessories, or having provisions for accessory stowage in its cover, and providing cushioning. The transit case is primarily a protective case intended to house the equipment and accessories but is not to be confused with a packing case, and is not intended for initial shipment without additional packing.

3.2.2 *Instrument case.*—An instrument case is a case protecting the instrument proper, and is part of the instrument. Thus, a dust cover, for example, may be part of an instrument case.

3.2.3 *Combination case.*—A combination

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case combines features of the transit case and the instrument case.

3.2.4 *Shipping case.*—The shipping case is the case or crate in which the complete test set (see 3.49) is shipped.

3.2.5 *Parts.*—Parts are defined as detail parts, such as resistors, capacitors, chokes, and sockets, and also as subassemblies, such as r. f. and i. f. amplifier sections, vibrators, voltage regulators, relays and terminal boards with parts mounted thereon.

3.3 Electronic standard parts, materials, and processes.—

3.3.1 Parts, materials, and processes shall be in accordance with specifications, other publications, and drawings, listed in section 2 as specified herein. Where practicable, values for the various parts shall be chosen from preferred listings as specified in applicable specifications and other publications listed in section 2.

3.3.2 *Selection.*—If more than one characteristic or tolerance of a part or material is permissible under the applicable specification, the broadest characteristic or widest tolerance consistent with the required performance of the test set shall be used. The design of the test set shall be such that it can accommodate any standard part having the maximum dimensions permitted for the type.

3.3.3 *Approval of nonstandard parts.*—Electronic parts not covered by existing standard specifications shall be satisfactory to the bureau or agency concerned. At the discretion of the bureau or agency concerned, samples of the proposed parts may be required to be furnished by the contractor for inspection and approval as to type. (See 4.2.2.)

3.3.4 If the bureau or agency concerned authorizes use of a nonstandard part in place of a standard part, the substitute part shall be mechanically and electrically replaceable by a standard part except in cases where the nonstandard part in question is of miniature or subminiature design. The contractor shall provide for mechanical replacement by allowing mounting space, providing mounting holes, and taking such other measures as may be necessary in the test set to permit replacement in the field with a standard part of the proper type. In such cases, the contractor shall inform the bureau or agency concerned of the type designation including nominal values and tolerances, of the standard part for which the nonstandard part is being substituted.

3.3.5 Miniature and subminiature parts satisfactory to the bureau or agency concerned shall be used wherever possible.

3.3.6 Mere convenience in design, procurement or shop processing shall not be regarded as justification for use of nonstandard parts and materials.

3.3.7 *Application of electronic standard specifications to re-orders.*—Whenever the test set to which this specification is applied has been manufactured previously using items not in accordance with specifications listed herein, the contractor shall use those items covered by applicable specifications which are electrically and mechanically interchangeable with items used in the test set as previously manufactured. When items in accordance with electronic standard specifications cannot be used because they will not be interchangeable with those previously used, or because they will require a major redesign of the test set, or because their use will not permit fulfillment of the test set performance requirements in the test set specification or contract, the contractor shall furnish evidence of such conditions to the bureau or agency concerned with a request for approval of the use of nonstandard items. Such a request shall be transmitted with substantiating data on electrical and mechanical details showing the comparison between the previously used items and the standard items which they most nearly approach in identity.

3.3.8 The requirements of this specification as to the use of electronic standard items and processes shall not relieve the contractor of the responsibility for complying with all test set performance requirements set forth in the specification or contract for the particular test set being furnished.

3.3.9 *Designation of electronic standard parts and materials.*—When it is required that manufacturers drawings, or specification data, or both be furnished to the bureau or agency concerned, for approval under this specification, the items indicated thereon shall be identified by reference both to the applicable electronic specification listed herein, and also to the

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proper class, grade, or type designation stipulated in that specification.

3.3.10 Detailed information relative to approved products listed on the Army-Navy-Air Force Qualified Products List, may be obtained from the cognizant Naval inspector for Navy purchases and from the contracting officer for Army and Air Force purchases.

3.4 Cables, connectors, and cable assemblies (interconnecting).—Cables, connectors, and cable assemblies (interconnecting) shall furnish satisfactory electrical connections and shall be so arranged and wired that no "hot" leads are terminated in pins or other exposed contacts which might be accidentally shorted or touched (see 3.42).

3.4.1 *Flexible power and interconnecting cable.*—Flexible power cables and interconnecting cable shall be in accordance with Specification MIL-C-915. For Army use, power input cable shall be an integral part of the equipment.

3.4.2 *Shielded cables.*—The shield shall be bonded to the metal shell of each connector used in the cable assembly, and shall be connected to a ground lead in the cable assembly.

3.4.3 *Coding.*—Leads in multiconductor cable assemblies and harnesses shall be color-coded to facilitate identification.

3.4.4 *Radio frequency cable.*—All types of radio frequency cables shall be in accordance with Specification JAN-C-17, and shall be selected from the Index of Army-Navy R-F Tranmission Lines and Fittings.

3.4.5 *Electrical connectors.*—Electrical connectors used on the combination case or instrument case shall be as follows:

3.4.5.1 *Connectors for radio frequency circuits.*—

3.4.5.1.1 Connectors, when required on test equipment, for maximum radio frequency lower than 3500 megacycles shall connect with plug UG-88/U as shown on Drawing RE49F246.

3.4.5.1.2 All connectors for maximum radio frequency higher than 3500 megacycles shall connect with plug UG-21B/U as shown on Drawing RE49F402 unless otherwise specified.

3.4.5.2 *Connectors for external primary power input circuits.*—

3.4.5.2.1 Alternating current power input connections shall be made by use of a recessed parallel-blade male convenience receptacle large enough to accept a female connector having a body diameter of 1⅜ inches except as specified in 3.4.1.

3.4.5.2.2 Direct current power input connections shall be made by use of a recessed right angle blade polarized male convenience receptacle large enough to accept a female connector having a body diameter of 1⅜ inches except as specified in 3.4.1. The radial blade shall be positive.

3.4.5.3 *Audio frequency circuit jacks.*—

3.4.5.3.1 *Microphone jacks.*—Microphone jacks shall be type JJ-033 in accordance with Specification JAN-J-641.

3.4.5.3.2 *Headset jacks.*—Headset jacks shall be type JJ-034 in accordance with Specification JAN-J-641, unless circuit switching jacks are required.

3.4.5.3.3 Where circuit-switching jacks are required, such jacks shall accommodate the following plugs:

Nomenclature:	Specification
PJ-068 3-contact	JAN-P-642
PJ-055B 2-contact	JAN-P-642

3.4.5.4. *Connectors for voltmeter or milliammeter measurements.*—Connectors for voltmeter or milliammeter measurements other than type PJ-055B shall accommodate test prods as shown on Drawing SC-C-6878 and SC-D-6883.

3.4.5.5 *Connectors for use with other equipment being tested.*—Connectors supplied with the test set for use on other equipment being tested shall conform to Specification AN-T-85. Conflicts between this paragraph and the general connector requirements shall be resolved by the bureau or agency concerned.

3.4.5.6 *Miscellaneous connectors.*—Connectors for other than voltmeter, milliammeters, alternating or direct current power input connections, radio-frequency circuits and microphone and headset cord circuits shall be in accordance with Specification MIL-C-5015. Contacts shall be so wired that there will be a minimum voltage difference between adjacent contacts.

3.5 Waveguides and waveguide assemblies.

3.5.1 Rigid waveguides shall be in accordance with Specification JAN-T-85 and shall be selected from the Index of Army-Navy R-F Transmission Lines and Fittings.

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3.5.2 Flexible waveguide assemblies shall be in accordance with Specification JAN-W-287 and shall be selected from the Index of Army-Navy R-F Transmission Lines and Fittings.

3.6 Capacitors.

3.6.1 *Electrolytic capacitors.*—The use of electrolytic capacitors in a particular test set shall be subject to specific approval by the bureau or agency concerned, based on detailed information submitted by the contractor as to their necessity, proposed use, and operating conditions in the test set. Where dry electrolytic polarized capacitors are used, they shall conform to Specification JAN-C-62.

3.6.2 *Fixed mica dielectric capacitors.*—Fixed mica dielectric capacitors shall be in accordance with Specification JAN-C-5.

3.6.3 *Fixed paper dielectric capacitors.*—Fixed paper dielectric capacitors shall be hermetically sealed in metallic cases and shall be in accordance with Specification JAN-C-25.

3.6.4 *Temperature compensating fixed ceramic dielectric capacitors.*—Temperature compensating fixed ceramic dielectric capacitors shall be in accordance with Specification JAN-C-20.

3.6.5 *Fixed capacitors, mounting.*—Fixed capacitors shall be securely mounted. They shall not be mounted by means of wire leads, except that unit-molded capacitors whose weight does not exceed ½ ounce may be secured by leads only, provided that the total length of leads from the capacitor to the holes in the terminal board or notches or holes in the terminals to which the leads are secured (whichever are the shorter) shall not exceed one inch, unless other mechanical support for the body of the capacitor is provided.

3.6.6 *Air dielectric capacitors.*—Nominal spacing between opposing plates shall be not less than 0.012 inch and actual spacing not less than 0.008 inch, unless a smaller nominal spacing is approved by the bureau or service concerned. Where nominal spacing less than 0.012 inch has been approved, the capacitors shall be enclosed in a dustproof case. Plates of all capacitors shall be free from grease, dust, dirt and metallic burrs. Grease and other foreign matter shall be removed from the plates by cleaning the assembled capacitor thoroughly with a suitable solvent. Rotors shall make low-resistance contact with connectors. As they are installed in the test set, all capacitors shall withstand without breakdown at least 500 volts, r. m. s., 60 cycles, between opposing plates. For variable capacitors this shall hold for all relative positions of rotor and stator plates. The rotors of all variable air dielectric capacitors shall be adequately supported.

3.6.6.1 *Air dielectric trimmer capacitors.* Air dielectric trimmer capacitors shall be in accordance with Specification JAN-C-92.

3.6.7 *Spring plate type variable capacitors.*—Spring plate type variable capacitors shall not be used.

3.6.8 *Ceramic-dielectric, variable capacitors.*—Ceramic-dielectric, variable capacitors shall be in accordance with Specification JAN-C-81.

3.6.9 *Paper-dielectric, fixed (nonmetallic cases) capacitors.*—Paper-dielectric, fixed (nonmetallic cases) capacitors shall be in accordance with Specification JAN-C-91.

3.7 Controls.—The characteristics of each control shall be such as to provide the desired control action under all specified service conditions.

3.7.1 *Stability.*—All tuning and operating controls of the test set shall be so designed that the setting, position, or adjustment of any control shall not be altered when the test set is subjected to the specified service conditions, including the tests of this specification. Locks shall be so designed as to preclude the shifting of controls when the locks are used.

3.7.2 *Ease of adjustment.*—The various controls shall rotate freely and smoothly without binding or excessive lost motion. Controls shall be adequately lubricated where necessary, and such lubricants shall not interfere with satisfactory performance of the test set. (See 3.41.3.)

3.7.3 *Ruggedness.*—All switches, levers, and other controls which are manipulated during operation of the test set shall be of such rugged design and construction that they will not be damaged when repeatedly operated by unskilled personnel under the specified service conditions. Where stops are used they shall be sufficiently rugged to prevent damage to the mechanism. All control knobs and handles shall be firmly secured to their control shafts

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by the use of set screws wherever that type of fastening is applicable. The set screw applications shall be in accordance with 3.21.3. Rotary switches shall be capable of withstanding a life test of 5000 cycles of make and break contact.

3.7.4 *Switches.*—All switches shall have a positive mechanical index locating each contact position except momentary on contact positions. Contacts shall be self-cleaning. In rotary switches the indexing mechanism shall be designed to prevent the movable element from coming to rest between contact positions. Insulation employed on rotary switches shall be either nonporous ceramic material or plastic material conforming to type LTS-E 4 of Specification JAN-P-13.

3.7.5 *Accessibility.*—Controls necessary for the operation of the test set shall be readily accessible, suitably arranged and of such size and construction to permit convenience and ease of operation under all service conditions including use of arctic mittens by operating personnel. Operating controls shall be so grouped and marked as to provide for simplicity and ease of operation.

3.8 **Control shafts.**—Mechanical linkages and splines (flexible control shafts) for external use shall be tuning shaft MC-215, in accordance with Specification AN-S-61.

3.9 **Crystals.**

3.9.1 *Piezo-electric crystal units.*—Piezo-electric crystal units shall be as specified in the individual test set specification.

3.9.2 *Rectifier crystals.*—Rectifier crystals shall be in accordance with Specification JAN-1. Rectifier crystal types not specifically designated in the individual test set specification shall be selected by the contractor from the Preferred List of Army-Navy Electron Tubes.

3.9.3 Circuits in test sets employing crystals, shall be designed so that the capacitance of the crystal circuit is equivalent to the capacitance conditions specified in the respective crystal specification.

3.10 **Dynamotors.**

3.10.1 Single output dynamotors shall be in accordance with Standard C75.13.

3.10.2 Multiple output dynamotors when used shall be of a quality consistent with Standard C75.13 and the test requirements of such dynamotors shall be, so far as applicable, in accordance with that standard.

3.11 **Electrical tape.**—Fabric or textile, pressure sensitive (adhesive or "friction") tape shall not be used except as noted in 3.14.10. Polyvinylchloride tape may be used for mechanical purposes when satisfactory to the bureau or agency concerned.

3.12 **Fasteners, snapslides, and studs.** Fasteners, snapslides, and studs shall be designed to fasten easily, and hold firmly and positively the movable or separable part to its mounting or other supporting structure under all service conditions. The designs shall be such as to permit quick and easy fastening or unfastening without requiring the use of any special tools. If snapslides and studs are used they shall be as shown on Drawings AN3195, AN3196, AN3197, AN3198, AN3199 and AND10082.

3.13 **Transformers, inductors, and choke coils.**

3.13.1 *Transformers and inductors (audio and power).*—Unless otherwise specified in the individual test set specification, transformers and inductors (audio and power) shall be in accordance with grade I of Specification MIL-T-27.

3.13.2 *Transformers, intermediate and radio frequency (Navy purchases only).*—Unless otherwise specified, intermediate and radio frequency transformers shall be in accordance with Specification 16T36.

3.13.3 *Choke coils (Navy purchases only).*—Unless otherwise specified in the individual test set specification choke coils shall be in accordance with Specification 16C38.

3.13.4 *Inductors, radio frequency.*

3.13.4.1 Inflammable material shall not be used in the construction of radio-frequency chokes, coils, and transformers.

3.13.4.2 *Impregnation.*—Impregnation of all multilayer windings is required, and also on laminated phenolic forms or on forms of any other material not grooved to hold the wire. Impregnation shall be evenly applied.

3.13.4.3 *Stability of "Q".*—The design of a coil shall be such that the "Q" of the coil within the operating frequency range shall not vary more than 10 percent as a result of immersion in a saturated solution of sodium chloride at

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20° C. (68° F.) for a period of 24 hours, followed by a rinse with tap water and drying for one hour in an atmosphere of 50° C. (122° F.) with a relative humidity of 30 percent.

3.13.4.4 *Dielectric strength.*—Inductors shall withstand 1-minute application of an alternating voltage between windings and between each winding and ground, all neighboring metal parts being grounded. The test voltage shall have a frequency not greater than 100 cycles per second and an r. m. s. value equal to three times the value of the maximum peak operating voltage between any two windings or between any winding and neighboring metal parts or ground. Where the maximum peak operating voltage is greater than 700 volts, the r. m. s. value of the test voltage shall be 1000 volts greater than 1.5 times the value of the maximum peak operating voltage.

3.13.4.5 *Variable inductors.*—Where a roller or slider is used in contact with the conductor, provision shall be made to insure firm low resistance contact and to limit the travel of the roller or slider to prevent its leaving the conductor.

3.14 Insulating and dielectric materials.

3.14.1 *Ceramic dielectric materials.*—Ceramic dielectric materials shall be in accordance with Specification JAN-I-12.

3.14.2 *General.*—Where ceramic parts are used for electrical insulation, all surfaces shall be glazed, or if glazing is impracticable, the surfaces shall be treated with a silicone compound (Dow-Corning DC 200 or equal) or an equivalent satisfactory to the bureau or agency concerned, except that surfaces not under electrical stress need neither be glazed nor treated with that compound. Ceramic parts shall not be treated with wax.

3.14.3 *Buffer washers for ceramic and vitreous surfaces.*—Buffer washers of suitable metallic or fibrous material shall be used between the otherwise facing surfaces of a ceramic or vitric insulator or vitreous enamel resistor and a metal part wherever practicable. Buffer washers of soft copper or other suitable metal shall be used wherever the heat resulting from soldering or internal heat from the normal operation of the test set is of such a degree as to cause deterioration of fiber washers in any manner.

3.14.4 *Impregnating and potting compounds.*—A compound shall not, either in the state of its original application or as the result of aging, have any injurious effect upon the insulation which it is designed to protect and shall not cause corrosion or deterioration of any adjacent parts. The compound shall not crack at temperatures above −65° C. (−85° F.) nor flow at temperatures below 85° C. (185° F.) unless contained in such a manner that the compound will not flow from the container.

3.14.5 *Laminated thermosetting plastic materials, sheets and plates.*—Laminated thermosetting plastic materials in the form of sheets and plates shall be in accordance with Specification MIL-P-997, JAN-P-13, or 17P25. Parts made of such materials, when used for electrical insulating, except types LTS-E-4, LTS-E-5, LTS-E-6, GMG, and GSG shall be vacuum impregnated with a suitable varnish and baked dry after all machining and punching operations are completed. The preferred base is glass cloth.

3.14.6 *Molded thermosetting plastic materials.*—Molded thermosetting plastic materials shall be in accordance with Specification JAN-P-14. When used for electrical insulation, parts fabricated of such materials, except transparent plastics and parts molded in final form without subsequent machining shall be vacuum impregnated with a suitable varnish and dried after all machining and punching operations are completed.

3.14.7 *Cast thermosetting plastic materials.*—Cast thermosetting plastic materials shall be in accordance with Specification JAN-P-77.

3.14.8 *Laminated thermosetting plastic materials, rods, and tubes.*—Laminated thermosetting plastic materials in the form of rods and tubes shall be in accordance with Specification JAN-P-79. Such materials, when used for electrical insulation except transparent plastics shall be vacuum impregnated with a suitable varnish and baked dry after all machining and punching operations have been completed.

3.14.9 *Rigid thermoplastic materials.* — Rigid thermoplastic materials shall be in accordance with Specification 17P8—Specification 72-78.

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3.14.10 *Cotton and linen.*—See 3.36.3 for use of cotton other than insulating material.

3.14.10.1 No insulated wire using cotton or linen in its construction shall be used in the test set, except as follows:

a. When the wire is used for coils on forms provided the insulation on the wire is completely sealed off against the atmosphere.

b. When the cotton or linen construction has been saturated with a moisture and fungus-resistant varnish in accordance with Specification JAN-C-173.

c. When the wire has been approved as in accordance with Specification JAN-C-76, except that in no case shall type WL wire be used for Navy purchases.

3.14.10.2 Cotton or linen shall not be used as the base or filler for laminated or molded plastic materials, unless the contractor definitely establishes to the satisfaction of the bureau or agency concerned that absorption of moisture will in no way cause degradation in performance of the test set as a whole, nor in any way adversely affect the particular part or any parts associated therewith or adjacent thereto. No untreated cotton-base or linen-base, laminated or molded plastic material shall be used for electrical insulation without the approval of the bureau or service concerned.

3.14.10.3 Cotton or linen shall not be used in the form of fabric or tape except as follows:

a. In construction of rotating electrical machinery where no other tape of sufficient mechanical strength is available.

b. On coils of parts such as inductors, transformers, and relays where the coil is subsequently completely impregnated and covered against moisture and fungus.

3.15 Insulators.

3.15.1 *Glass insulators.*—Glass insulators shall be in accordance with Specification JAN-I-9.

3.15.2 *Glass bonded mica insulators.*—Glass bonded mica insulators shall be in accordance with Specification JAN-I-7.

3.15.3 *Porcelain insulators.*—Porcelain insulators shall be in accordance with Specification JAN-I-21.

3.15.4 *Steatite insulators.*—Steatite insulators shall be in accordance with Specification JAN-I-8.

3.15.5 *Permitted nonstandard ceramic insulators and forms.*—Permitted nonstandard ceramic insulators and forms shall be made of ceramic insulating materials in accordance with Specification JAN-I-10.

3.16 Locking devices.—Locking devices shall be employed with all screwthread assemblies.

3.16.1 *Lock washers.*—Lock washers shall be heat treated and plated phosphor-bronze, or corrosion-resisting or rustproofed steel. Lock washers shall not be used in direct contact with plastics and other soft materials. Lock washers shall be placed under the nuts, if used, rather than under the screwheads.

3.16.2 *Other devices.*—Self-locking nuts satisfactory to the bureau or agency concerned, safety wiring, and castellated nuts with cotter pinning are acceptable.

3.16.3 *Staking or retaining compounds.*—Staking by means of upsetting metal, or by using a retaining compound, is acceptable only when other methods are impracticable or unsatisfactory. Neither shall be used where the assembly is frequently disassembled.

3.16.4 *Nonmetals.*—Where screw threads engage tapped holes in nonmetals, a suitable retaining compound shall be applied to such threads. Tapped holes shall be thoroughly covered with a suitable retaining compound. Tapped holes shall not be employed in plastic materials, except in applications where loosening of a screw will not cause damage to or maloperation of the test set. When screw heads are adjacent to nonmetals. with or without washers, the facing surfaces shall be thoroughly covered with a suitable retaining compound. Threaded metal inserts shall be used where added strength is necessary or where the screw will be removed and replaced frequently in service.

3.17 Magnesium alloys.—When magnesium alloy parts are used in direct contact with dissimilar metals, extreme caution shall be exercised in the selection of the proper alloy to guard against electrolytic corrosion. Insulating coatings such as zinc chromate primer shall be applied or, in cases where extreme exposure is contemplated, the use of gaskets is desirable.

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When tested as an assembly such magnesium parts and all other parts of dissimilar metal normally attached to them shall successfully withstand a 96-hour salt-spray test in accordance with Specification QQ-M-151. At the completion of the salt-spray test the samples tested shall exhibit no signs of excessive pitting or corrosion, and shall be capable of easy disassembly.

3.18 Meters and electrical indicating instruments.

3.18.1 Meters employed in test equipment shall be sealed meters in accordance with Specification JAN-I-6. Meters shall be properly bonded to the panel and shall use a conducting glass window or coating material where necessary to comply with the noise suppression requirements.

3.18.2 Where clear plastic materials are used for instrument windows they shall be in accordance with Specification JAN-P-80.

3.19 Relays.—The number of relays and contacts thereon shall be held to a minimum consistent with the required performance.

3.19.1 Relay contacts shall be of the self-cleaning and noncorrosive type, wherever practicable.

3.19.2 Relays shall be hermetically sealed or shall be treated or constructed to operate satisfactorily during 30 days of humidity cycling, as specified in 4.4.2, with rated coil voltage applied between the coil and frame but with no operating current flowing.

3.19.3 Relays energized from the primary power supply shall pull in and make firm and positive contact when the reference voltage of that supply is reduced to 68 percent, and shall not drop out when the reference voltage of that supply is reduced to 50 percent. For the purpose of this test, the reference voltages shall be as follows:

Nominal supply voltage:	*Reference supply voltage*
26 volt d. c.	26.5 volt d. c.
13.5 volt d. c.	13.75 volt d. c.
6.65 volt d. c.	6.75 volt d. c.
115 volt a. c.	115 volt a. c.
220 volt a. c.	220 volt a. c.

3.19.4 The operational characteristics of the other relays shall be as specified in the individual test set specification, and as specified in 3.48.

3.20 Resistors.—The rating and placement of a resistor in a particular test set shall be such that the permitted maximum temperature at any point on the surface of the resistor shall not be exceeded under the highest ambient temperature specified as a service condition for the complete test set, and with any type of operation provided by the test set.

3.20.1 *Composition resistors.*

3.20.1.1 *Fixed.*—Fixed composition resistors shall be in accordance with Specification JAN-R-11.

3.20.1.2 *Variable.*—Variable composition resistors shall be in accordance with Specification JAN-R-94.

3.20.2 *External voltmeter resistors.*—External voltmeter resistors shall be in accordance with Specification JAN-R-29 or JAN-R-93, as applicable. Such resistors may also be used in other high voltage applications for which they may be found suitable.

3.20.3 *Tapped resistors.*—Fixed and variable resistors having fixed taps may be employed, where required, by overall design; but their use shall be held to a minimum.

3.20.4 *Wirewound resistors.*

3.20.4.1 *Fixed.*—Fixed wirewound resistors shall be in accordance with Specification JAN-R-26. Characteristics G and J resistors may be used only in pressurized or hermetically sealed test sets.

3.20.4.2 *Variable.*—Variable wirewound resistors shall be in accordance with Specification JAN-R-19.

3.20.4.3 *Precision.*—Precision wirewound resistors shall be in accordance with Specification JAN-R-93.

3.20.4.4 *Fixed (low power).*—Fixed wirewound low power resistors shall be in accordance with Specification JAN-R-184.

3.20.4.5 *Rheostats.*—Power type wirewound rheostats shall be in accordance with Specification JAN-R-22.

3.21 Screws and bolts.—Screws and bolts in sizes smaller than No. 4 shall not be used on parts other than meters, relays, circuit breakers and other small precision assemblies and on miniature tube sockets, except that smaller screws may be used on parts which are not intended to be taken apart and when the design requirements cannot otherwise be met. In any

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case sizes smaller than No. 6 shall preferably not be used for mounting parts or assemblies on panels, brackets and the like.

3.21.1 *Screw lengths.*—The length of screws and bolts shall provide a minimum of 1½ threads and a maximum of 1½ threads plus ⅛ inch length beyond the nut or joining part, except where such projection will result in corona discharge or where design requirements cannot otherwise be met.

3.21.2 *Thread engagement.*—Thread engagement of screws and bolts shall provide a minimum of ⅛ inch effective thread in tapped parts other than nuts when the assembly is frequently disassembled. A minimum of three complete threads shall be used in aluminum, aluminum alloys and other soft materials or nonmetals. A minimum of two complete threads shall be used in stronger materials. This paragraph is not applicable to the use of sheet spring nuts in accordance with Specification AN-N-12 or instrument mounting nuts in accordance with Specification AN-N-11, nor to other nut devices specifically approved by the bureau or agency concerned as being comparable with the designs covered by those two specifications.

3.21.3 *Set screws.*—All set screws shall be hardened slotted-head or Allen head type. Where it is not necessary to have a round shaft, all shafts provided with a knob shall be flatted. One set screw may be used on a flatted shaft. Two set screws at 90 to 120 degrees shall be used where the shaft is not flatted. All set screws used in the test set shall have one type of head. Cone-point set screws shall not be used except where the opposing material has been properly countersunk to receive the cone point. (See 3.29.2.)

3.21.4 *Flat-head screws.*—Flat-head screws countersunk in sheet or thin material, the thickness of which is less than one and one-half times the height of the head of the screw, shall not be used. Wherever flathead screws are used the countersinking shall be such that the screw head is properly and completely seated in the countersunk material. The use of dimples on thin sheet metal, when using flat-head screws, will be permitted.

3.21.5 *Self-tapping and sheet metal screws.*—Where no other suitable means exist, self-tapping and sheet metal screws may be used and shall be in accordance with Drawings AN504, AN506, AN530 and AN531, as applicable, and Specification AN-S-24. The use of self-tapping screws in test equipment is subject to the limitations on use which are noted in Specification AN-S-24 except that a combination of sheet metal screws in accordance with Drawings AN530 and AN531 and sheet spring nuts in accordance with Specification AN-N-12, or instrument mounting nuts in accordance with Specification AN-N-11, or specifically approved comparable self-locking fastening devices is approved where the only limitation on use noted in Specification AN-S-24 is the frequency of removal in the process of servicing the equipment.

3.21.6 *Sheet spring nuts.*—Sheet spring nuts shall be in accordance with Specification AN-N-11 or AN-N-12. These types of nuts shall not be used unless approved for each application by the bureau or agency concerned. They shall not be used in the primary structural assembly of a major or minor assembly nor on screws used to mount such articles of equipment where failure or loosening of one or more of the associated mounting screws would result in damage to or failure of the test set. They shall not be used in assemblies where the initial tension, due to tightening the nut on the screw, may be relieved by external forces on the assembly or by shrinkage of the materials used in the assembly.

3.22 **Screw threads.**—All threads shall be in accordance with Handbook H28.

3.22.1 For screws, bolts, nuts, and similar devices, the threads shall be of the National coarse thread series unless used for adjustment where they may be of the National fine thread or National extra fine thread series.

3.22.2 *Commercial threads.* — Commercial threads in general use for mounting variable resistors, toggle switches, and similar devices are acceptable.

3.22.3 *Threads in soft metals.*—Aluminum and other soft metals shall not be threaded for use as parts that are frequently disassembled.

3.22.4 *Aluminum anti-seize compound.*—Aluminum anti-seize compounds shall be used in all threaded aluminum-alloy assemblies, ex-

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cept those involving machine screws or self-tapping screws only. The compound shall be in accordance with Specification JAN-A-669 or AN-P-51, or shall have equal or greater anti-seize properties and at least equivalent characteristics in all other respects

3.22.5 *Nonmetals.*—Where screw threads engage tapped holes in nonmetals, such tapped holes shall be provided with metallic inserts secured by knurling or other method satisfactory to the bureau or agency concerned, which will preclude their movement.

3.23 **Shock and vibration insulators.**—Devices such as vacuum tubes or other parts whose life or dependability would be adversely affected by shock and vibration may be individually protected from shock and vibration in any direction. In cushioning, all mountings shall be of such material as not to require replacement during an extended period of operation, and shall be so designed and installed as to be readily replaceable without major disassembly. If cushioning is employed sufficient clearance shall be provided between parts to preclude the possibility of a cushioned part striking any other part under any specified conditions of shock or vibration.

3.24 **Soldering.**—Soldering shall be so performed that both a good electrical and a good mechanical connection are assured.

3.24.1 *Mechanical assemblies.*—Unless specifically approved by the bureau or agency concerned, no assembly shall depend solely on soft solder for mechanical strength except for variable capacitor plates and sections and other relatively light parts that are of accepted commercial design, and that have by actual use, proved to be generally suitable for use in electronic equipment.

3.24.2 *Damage.*—Soldering shall be so performed that no damage is caused to neighboring parts.

3.24.3 *Electrical connections.*—All joints shall be mechanically secure before soldering. No connection shall depend on soft solder alone for mechanical strength.

3.24.4 *Cleaning.*—All joints shall be clean before the application of solder.

3.24.5 *Cold joints.*—There shall be no evidence of "cold joints". Sufficient heat shall be applied so that solder flows freely and completely over the joint.

3.24.6 *Excess solder and flux.*—The soldered connections shall be free from excess solder and flux such as drippings and joints. Generally only the amount of solder and flux necessary to solder the joint shall be applied.

3.25 **Solder and soldering flux.**—Only composition Sn70 or Sn60 solder in accordance with Specification QQ-S-571 shall be used unless specific authority is granted by the bureau or agency concerned for the use of any other class. Only flux which is noncorrosive and which will not support the growth of fungus may be used in the assembly of test equipment or any part thereof.

3.26 **Springs.**—The type and grade of material used and the processing of the material shall be entirely suitable for the intended purpose.

3.26.1 *Contact springs.*—Beryllium-copper, phosphor-bronze, coin silver or nickel-silver, tempered and plated as necessary, shall be used for contact springs.

3.26.2 *Tempering.*—Springs of beryllium-copper alloys shall be heat-treated to the proper spring temper after forming. Springs made from materials which are not heat treatable shall have the direction of the grain within 45 degrees of the linear length.

3.27 **Switches.**

3.27.1 *Sensitive switches.*—Sensitive switches shall be in accordance with Specification JAN-S-63.

3.27.2 *Toggle switches.*—Toggle switches shall be in accordance with Specification JAN-S-23.

3.27.3 *Vacuum switches.*—Vacuum switches shall be in accordance with Specification JAN-S-57.

3.28 **Terminals.**

3.28.1 *Terminals attached to parts.*—Terminals attached to parts shall be mechanically strong, and shall not bend or break when soldering or unsoldering attached connections. Locking terminals are not acceptable in lieu of lock-washers. Terminals of potted parts shall be so designed and fastened to the terminals insulating strip or plate (or to the enclosure itself if this arrangement satisfies the insulation requirements) as not to cause any degradation in

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the moisture-excluding property of the enclosure by the normal soldering and resoldering of external leads to the terminals. Terminals shall be so spaced as to assure a leakage resistance sufficiently high, and to insure safe creepage distances for the particular application under specified service conditions of high humidity or high elevation. In general, the creepage distances and mechanical spacings specified in Specification MIL-C-5015 shall be considered satisfactory.

3.28.2 *Terminals attached to wires.*—Terminals attached to wires may be of either the solder or solderless type. The solderless terminals shall be as shown on Drawing AN-659 or in accordance with the requirements for WT type terminals of Specification 17C13.

3.28.3 *Terminal strips and boards.*—Terminal strips and boards shall be made of ceramic material or of low-moisture-absorption arc-resistant plastic material. Adequate terminal spacing or barriers shall be employed to prevent breakdown or low leakage resistance under high humidity, including condensation, or high altitude conditions. The creepage distances and mechanical spacing specified in Specification MIL-C-5015 shall be considered satisfactory. If unglazed ceramic material is employed the back of the board shall be thoroughly coated with the class of varnish applicable to the particular equipment in accordance with Specification JAN-C-173. After the terminals have been affixed to the board, the front of the board shall be also similarly coated unless a thorough coating will be accomplished subsequently with an overall spray treatment. Coating shall not be applied to arc-resistant plastics and glazed ceramics employed on terminal strips and boards.

3.29 Special tools.

3.29.1 The design of test equipment shall be such that the need of special tools for tuning, adjusting or servicing shall be kept to a minimum and the use of such special tools shall be subject to the approval of the bureau or agency concerned. Special tools required for operational adjustments shall be mounted securely in the test set, in a convenient place.

3.29.2 *Set-screw wrenches.*—One wrench for each size of set screw used shall be mounted securely in a convenient place in the test set.

3.30 Tubes, electron.

3.30.1 Electron tubes shall be in accordance with Specification JAN-1.

3.30.2 The number of tubes and types of tubes used in each particular test set shall be kept to a minumum. Electron tubes used in the design of each particular test set shall be chosen from the Preferred List of Army-Navy Electron Tubes. Use of any tubes not shown on the preferred list will be acceptable only if satisfactory to the bureau or agency concerned. Before consideration will be given to acceptance of a nonpreferred type, the following information is required:

a. Quantitative data on the electrical advantages of the proposed tube over the nearest equivalent type on the preferred list.

b. Proof that the proposed tube is mechanically adequate for the service intended.

c. Submission of tube test data and performance limits of the tube which would have to be maintained by the tube manufacturer to insure satisfactory performance of the test set.

d. Information on production availability of the tube.

3.30.3 The contractor is not authorized to develop any new tubes necessary for the test set, but is required to notify the bureau or agency concerned should the need for such development arise.

3.30.4 *Design requirements.*—Test equipment shall be so designed that the specified performance will be obtained from tubes on the Preferred List of Army-Navy Electron Tubes, meeting and operating within the requirements of Specification JAN-1. Test sets and the electron tube circuits therein shall be so designed that, when a tube with any practical combination of tube characteristics controlled by Specification JAN-1 is used (all other tubes being average), the test set will meet the performance standards specified in the individual test set specification. In general, the operation of a circuit shall not depend on an electron tube characteristic not controlled by Specification JAN-1.

3.30.5 *Design information.*—The contractor shall supply any detailed design information

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concerning the specific application of electron tubes that may be requested by the bureau or agency concerned.

3.30.6 *Tube complement reports.*—All tube complements, even though tentative, shall be reported by the contractor to the bureau or agency concerned as soon as determined; all changes in such tube complements shall also be reported as they occur, during the course of the contract. The bureau or agency concerned shall be advised promptly of the final tube complement as soon as it has been determined.

3.31 Tube sockets.

3.31.1 *Miniature tube sockets.*—Miniature electron tube sockets shall be in accordance with Specification JAN-S-28.

3.31.2 *Other tube sockets.*—Tube sockets shall be of the single unit type. Gang type sockets are not acceptable. This requirement is not intended to preclude the assembly of several tube sockets on a supporting frame so long as the contact clips for any tube are on insulating material which is discontinuous with that for any other tube. Sockets shall be approved by the bureau or agency concerned as to their ability to retain a tube under service conditions. The socket contacts shall be made of silver-plated beryllium-copper or silver-plated phosphor-bronze. Tube socket body material shall be one of the following:

a. Ceramic, in accordance with grade L-3 of Specification JAN-I-10 or better.

b. Low-loss molded plastic, in accordance with type MTS-E-4 of Specification JAN-P-14 or

c. Molded plastic, in accordance with type RTP-E-2 or RTP-EM-3 of Specification 17P8—Specification 72-78.

3.32 Welds.—All welds shall be of ample size, with thorough penetration and good fusion, and shall be substantially free from injurious scabs, blisters, abnormal pockmarks, bruises or other surface defects. Welds shall be capable of withstanding a strength test to destruction in which failure shall not occur within the weld itself, as determined by tests made whenever so required by the Government inspector on sample pieces representative of production and welded on the machines used in production. There shall be no evidence of harmful effect due to welding process on material adjacent to weld. All welds shall be thoroughly cleaned and the excess of flux shall be chemically neutralized and removed.

3.33 Wires.

3.33.1 Stranded conductors of suitable degrees of flexibility shall be used wherever practicable within the requirements of this specification. All wires shall be of such cross section and temper as to provide ample and safe current-carrying capacity and mechanical strength. Wire shall not be smaller than 0.0253 inch diameter (A. W. G. No. 22) and wires directly associated with the primary power circuit shall be of a size consistent with the ratings of the protective devices used. Insofar as is practicable all wiring in the test set shall be distinctly color-coded in order to facilitate testing and the location of faults. Hook-up wire used within the test set shall be in accordance with Specification JAN-C-76, except that type WL shall not be used for Navy purchases.

3.33.2 *Insulation on conductors.*—Insulation on conductors shall be integral with the conductor except that insulation of the sleeving type may be used where authorized by the bureau or agency concerned provided its length is not greater than 4 inches and is sufficient to prevent slippage along the conductor to such an extent as to expose a section of the conductor that should be insulated and provided that the internal surfaces of the sleeving and the contained wires shall be treated with a fungistatic agent. Insulation shall provide high dielectric strength and high leakage resistance when the test set is operated under the designated service conditions. Insulation thickness shall be not less than 0.02 inch. The mechanical and electrical properties of the insulating materials and impregnating compounds shall not be permanently impaired as a result of being exposed to extremes of temperatures from −65° C. (−85° F.) to +100° C. (+212° F.). Rubber without braid may be used only where the insulation is not depended upon to space or support the conductor or where a number of conductors are assembled in the form of cordage and the assembly provided with a braid covering suitably varnished or lacquered. All conductor insulation shall be of the noncombustible or the slow-burning type, that is with the insulated conductor held in a horizontal position

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in still air, self-sustained combustion of the insulation or lacquer shall not progress at a rate in excess of 2 inches in 30 seconds. Color-coding in insulation shall be fadeproof.

3.34 Wiring.

3.34.1 All connections within the test set shall be suitably supported so as to obviate breakage and minimize changes in frequency or output due to severe vibration and acceleration of shock encountered under the specified service conditions. Soldering of wires on shields to flat surfaces, without provisions thereon for mechanical security of the connection in addition to solder, is not acceptable. The use of a flexible and a solid conductor in a continuous run without a support at their junction is not acceptable. Solid wire not smaller than 0.0253 inch diameter (A. W. G. No. 22) may be used in all units of the test set that are shock mounted, provided that such wires are so insulated or held in place that they do not fail or show excessive motion when the test set is subjected to the vibration or shock encountered under the specified service conditions.

3.34.2 Before soldering, all wires connecting to terminals shall be securely fastened to the terminals by crimping the terminals firmly upon the wire or the wire upon the terminals, except in cases where the mechanical configuration of the wire or design of the terminal is such as to hold the wire securely in place without crimping. Unless otherwise indicated or specified for a detailed part, no wire connection shall depend solely upon solder for mechanical strength. In no case shall electrical connections be made by clamping wires smaller than 0.0641 inch diameter (A. W. G. No. 14) between metal parts. Such connections shall be soldered, or soldering lugs or terminals may be used in lieu thereof.

3.34.3 Whenever any bolts, screws, nuts, studs, or rivets are used in, or as a part of, a radio frequency circuit, all connections thereto shall be securely soldered, wherever practicable. Soldered connections are not considered practicable at studs of molded phenolic capacitors, meter terminals, relay contacts, and the like, and other places where damage from overheating will result.

3.34.4 In no case shall electrical connections depend upon wires, lugs, terminals and the like clamped between a metallic member and an insulating material of other than a ceramic or vitric nature. Such connections shall be clamped between metal members, preferably such as an assembly of two nuts, two washers and a machine screw. If such an assembly is not used, and maintenance of a tight connection depends upon the resistance of an insulating material of other than a ceramic or vitric nature to compressive stress or shear, such connections shall be securely soldered.

3.34.5 Wherever wires are run through holes in metal partitions, shields and the like less than ⅛ inch in thickness, the holes shall be equipped with suitable grommets for mechanical protection of insulation otherwise subject to abrasion. Panels ⅛ inch or more in thickness either shall have grommets or shall have the hole edges rounded to a radius equal to one-half the thickness of the material. Care shall be exercised in the running of hookup wire to insure that it is not carried over or bent around any sharp corner or edge which might in time cut through the insulation. In order to prevent deterioration of the conductor by heat, care shall be taken to insure that wiring is not exposed to local temperatures appreciably above ambient temperatures and that it cannot come in direct contact with heated parts.

3.34.6 Conductors shall be bound into a cable wherever possible and securely held by means of wraplock, cord, or other suitable means. All lacing cords shall be made of fungus-inert plastic, such as nylon or cellulose-acetate rayon, etc., or shall be treated to provide water repellent and mildew resistant characteristics and to pass the tests of Specification CCC-T-191, except that copper compounds shall not be used.

3.34.7 Conductors using metallic shielding unprotected by an outer insulation shall be so secured as to prevent the shielding from coming into contact with exposed terminals or conductors. Shielding shall be terminated at a suitable distance from the exposed conductor. The shield shall be bonded to the chassis.

3.34.8 *Flexible leads.*—Flexible leads shall conform to Specification 71-3310.

3.35 Finishes.—All parts of the test set not covered by subsidiary specifications and having protective finishes shall be resistant to corrosion and there shall be no destructive corrosion after

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a 100-hour salt-spray test in accordance with Specification QQ-M-151. Destructive corrosion shall be construed as being any type of corrosion which in any way interferes with mechanical and electrical performance. Bright finishes shall not be used on external parts. Where cleaning operations on metal parts are not specified in detail, they shall be in accordance with the best conventional commercial practices.

3.35.1 *Finishes of transit, combination and instrument case.*—External surfaces shall be finished with gray enamel conforming to Specification 52E14. Class 2 enamel shall be used if weather protected and class 1 enamel shall be used if weather exposed.

3.35.2 *Aluminum alloy.*—Approved processes for finishes of aluminum alloys are those commercially known as anodic, alumilite, caustic dip, sand blast, nasat, and supersat. The finish processes as specified in Specification AN-QQ-A-696 will be satisfactory.

3.35.2.1 *Bonded aluminum surfaces.*—Aluminum surfaces which are to be bonded shall not be processed without making provisions, suitable to the bureau or agency concerned, to insure a low resistance electrical path.

3.35.3 *Fasteners and assembly screws.*—All external fasteners and assembly screws which are manipulated, loosened or removed in normal processes of installing and servicing the test set shall be of such color as to provide strong contrast with the color of the surface upon which they appear. Other external fasteners and assembly screws shall be similar in color to the surface upon which they appear.

3.35.4 *Ferrous alloys.*—Corrosion-resisting ferrous alloys shall be given a passivation treatment but need not receive any other protective plating or finish unless such plating or finish is necessary or desirable for electrical or mechanical reasons. Ordinary iron and steel shall be plated or finished as specified in 3.35.7. The iron or steel laminations used in magnetic circuits need not be plated or given a protective finish if they are otherwise protected against corrosion.

3.35.5 *Internal parts.*—Black or gray paint or enamel or clear lacquer may be used on internal parts and interior surfaces of housings.

3.35.6 *Magnesium and magnesium alloys.*—Magnesium and magnesium alloys shall be finished in accordance with either type II, III, or IV application of Specification AN-M-12. (See 3.17.)

3.35.7 *Plating.*—Plating shall be in accordance with the applicable specification for each particular type as follows:

Cadmium	AN-P-61
Chromium	AN-P-39
Nickel	AN-P-34
Zinc	AN-P-32

3.35.8 *Zinc and zinc plated parts.*—All zinc parts and zinc plated parts shall be given a dichromate treatment such as Cronak (N. J. Zinc Co.), Iridite (Rheem Research Products, Inc.), or equal.

3.35.9 *Other standard finishes (for Army use only).*—Type I finishes in accordance with Specification 72-53 are approved as alternates to any differing requirements of 3.35 to 3.35.8 inclusive, except that the colors specified therein shall be adhered to.

3.36 Moisture and fungus resistant treatment.

3.36.1 *General.*—Moisture and fungus resistant treatment shall be applied in accordance with Specification JAN-T-152. The coating material shall be varnish, of the applicable class, in accordance with Specification JAN-C-173.

3.36.2 *Moisture and fungus-resistant treatment of power and miscellaneous connectors.*

3.36.2.1 *Connectors with multiple insulators.*—If a plug or receptacle includes two or more insulating bodies which face or fit together, the mating or facing surfaces shall be completely coated with insulating and sealing compound in accordance with Specification AN-C-128. This shall be performed prior to assembly of the plug or receptacle and is applicable to plugs and receptacles regardless of whether wired in the test set, supplied as individual articles of a complete test set or supplied as maintenance parts.

3.36.2.2 *Connectors wired in.*—All plugs and receptacles furnished as integral wired-in parts of test sets shall be processed as follows:

3.36.2.2.1 A piece of moisture and fungus-resistant plastic or glass-braid tubing, or equal shall be placed over each wire. The tubing shall be long enough to cover completely the

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terminal and at least ½ inch of insulation of the wire which is attached to the terminal. If bare wire is used, the tubing shall be long enough to extend at least ¼ inch beyond the terminal.

3.36.2.2.2 After all soldering and connecting of wires thereto have been completed, each soldered joint and wire including at least 1 inch of insulation shall be coated with a class 1 or class 2, as applicable, moisture and fungus-resistant varnish in accordance with Specification JAN–C–173. The tubing shall be pushed over the terminal and recoated with the applicable class of coating material.

3.36.2.2.3 Insulating and sealing compound in accordance with Specification AN–C–128, shall be applied under the following conditions:

a. To the front mating surfaces, other than electrical contact areas and coupling threads, of each combination of mating plug and receptacle internal to test set so that, when the plug and receptacle are mated the spaces within the plug and receptable are filled with the compound. The front surface of a receptacle on the exterior of test set shall not be so treated.

b. To the back of each plug or receptacle if the procedure specified in a. above has not resulted in covering that area.

c. During the last stage of final assembly.

d. With precautions taken to prevent the accumulation of dirt by the compound or smearing of the compound on any part of the test set.

3.36.3 *Cotton duck and webbing.*—Cotton duck and webbing shall be treated to be water repellent and mildew resistant.

3.36.3.1 *Water repellency.*—When tested in accordance with Specification CCC–T–191, section III, paragraph 5b (2), the material shall absorb not more than 25 percent water.

3.36.3.2 *Mildew resistance.*—The treated material shall show not more than 10 percent loss in breaking strength when tested in accordance with the 14-day test described in the supplement to Specification CCC–T–191, section VIII, paragraph 1b, following the leaching test described in section VIII, paragraph 1a (2). The mildew resistant treatment shall not be hygroscopic nor shall the material become tacky nor lose its normal flexibility due to the treatment. Where the duck or webbing is in contact with natural or synthetic rubber, treatments containing copper shall not be employed.

3.36.4 *Felt.*—Felt shall be suitably treated to be water-repellent and fungus resistant to meet the type III treatment of Specification 100–46.

3.36.5 *Leather.*

3.36.5.1 All leather shall be treated for fungus resistance and water repellency with the following solution or equal:

	By weight percent	Per gallon ounces
Salicylanilide	2	2.5
Hexalin (cyclohexanol)	25	32.0
Refined paraffin wax (M. P. 49° C. (120.2° F.) to 55° C. (131° F.)	10	13.0
Stoddard solvent	63	80.5

3.36.5.2 *Directions.*—Dissolve the salicylanilide in the hexalin. Add the paraffin wax and stir until dissolved. (Heating to 38° C. (100.4° F.) will help.) Add the Stoddard solvent with constant stirring. Use at 22° C. (71.6° F.) or room temperature whichever is higher. Solution should never be used below 22° C. (71.6° F.) and must be free from visible undissolved particles of wax. Heating the solution should be employed only in cases where it is necessary to maintain a minimum of 22° C. (71.6° F.) solution temperature. Solutions that have been chilled or frozen to the point of precipitation of the wax should not be used until they have been warmed to at least 22° C. (71.6° F.) and sufficient time allowed (plus stirring) for the wax to be redissolved. Caution: Waxes of higher melting points than specified or those generally known as microcrystalline should not be used. For the available supply source of salicylanilide see 6.3.

3.36.5.3 *Processing procedure.*—The leather shall be cleaned with a dry cloth or brush and should be in a dry condition before treatment. The leather shall be immersed in the solution for 5 minutes, and then dried thoroughly. If facilities are available, the leather may be dried at 49° C. (120.2° F.) for 8 hours or 60° C. (140° F.) for 5 hours.

3.37 Nomenclature.—The complete test set will be assigned Army-Navy nomenclature,

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which comprises a standard catalog name and a type designation. Any nomenclature which is not completely given in the contract or in the individual specification for a particular test set will be furnished by the bureau or agency concerned upon written application by the contractor, who shall furnish detail descriptive information in accordance with Publication JANP 109.

3.38 Marking and identification.

3.38.1 *Name plates.*

3.38.1.1 Where space permits, name plates shall be 3 inches long by 1 inch wide as shown on figure 1. If a smaller size is indicated, the name plates shall be as large as possible, and shall be similar to figure 1 in arrangement and content.

3.38.1.2 The lettering and blocks in the respective lines shall be substantially of the style and size shown on figure 1 and shall be light colored on a dark background. The information to appear shall include the following in the indicated order from top to bottom. Additional pertinent lines of information, as applicable, may be inserted above, below and between the following minimum lines of information, subject to the provisions of 3.38.1.3:

a. The complete Army-Navy nomenclature (name and type designation).

b. The external supply voltage, frequency and phases, if applicable.

c. A block approximately 1 by ⅛ inch suitable for stamping in a number having a maximum of twelve characters.

d. The contract number and serial number, with a circular block in the center of the line, between the numbers; numbers shall be followed by a colon and manufacturer's designating letters or code.

3.38.1.3 All name plate drawings shall be submitted to the bureau or agency concerned for approval. Approval must be obtained prior to the fabrication of name plates.

3.38.1.4 Serial numbers and manufacturer's designating letters or code will be assigned by the bureau or agency concerned.

3.38.1.5 *Laminated thermosetting plastic materials for name, graphic chart, instructions, and designating plates.*—Laminated thermosetting plastic materials for name, graphic chart, instructions, and designating plates shall be in accordance with Specification JAN-P-78.

3.38.2 *Type designations and marking.*

3.38.2.1 The name plate of the test instrument proper (single principal item of the test set) shall carry the complete Army-Navy nomenclature of the test set as a whole. The use of such nomenclature shall be reserved for principal items.

3.38.2.2 Each accessory and transit case supplied shall have its type designation marked on its name plate, or shall be otherwise appropriately marked.

3.38.2.3 If a transit case is used, its name plate shall be on the inside of the cover. The title on the name plate shall have no reference to the Army-Navy nomenclature of the test set of which the principal item is housed in the case.

3.38.2.4 On the inside of the cover of the transit or combination case shall be marked the appropriate equivalent of "The following are part of and included in the _____."
(Army-Navy Nomenclature)
The complete Army-Navy nomenclature for the accessories and the transit case, if any, and a list of operating maintenance parts shall follow. The quantity and type of any batteries required shall be listed. In the event that the case includes no item with Army-Navy nomenclature, the nomenclature of the group shall appear at the top of the list.

3.38.2.5 The Army-Navy nomenclature of the complete test set shall be stenciled on the exterior of each transit and combination case on at least three adjacent surfaces in black block letters as large as possible, but not exceeding 2 inches in height.

3.38.2.6 If there are two or more cases (transit or combination) for the test set the above procedure shall be used with the following additions:

a. The Army-Navy nomenclature of the complete test set shall be stenciled on the exterior of each of the cases.

b. There shall be marked on the exterior of each case that the case is "Case n of m", where "n" is the particular case number and "m" is the total number of cases in the test set.

c. The contents of all the cases shall be

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listed by case number on the inside of the lid of each case.

3.38.3 *Other markings.*—All assemblies, subassemblies and parts which are likely to be replaced in service use shall be marked and identified with information in accordance with 3.38.3.1, 3.38.3.2, 3.38.3.3, 3.38.3.4, 3.38.3.5, and 3.38.3.6, and using the methods specified in 3.38.4.

3.38.3.1 The reference symbol used in the Table of Replaceable Parts in the Handbook of Maintenance Instructions Technical Manual or Instruction Book as applicable, shall be marked where practicable on the supporting structure adjacent to each replaceable part in such a position as to be readily viewed when tubes, access doors, or shields are removed. Reference symbols may be omitted from the supporting structure in the case of small resistors and capacitors that are supported by their wire leads and when the test set would otherwise have to be increased in size to provide identification.

3.38.3.2 *Plug receptacles.*—The visible surface adjacent to the plug receptable shall be marked with the Army-Navy type designations of the cables which attach to the plug.

3.38.3.3 *Tube sockets.*—The tube side of the chassis adjacent to the tube socket shall be marked with the JAN type designation of the tube and the reference symbol used in the Handbook of Maintenance Instructions Technical Manual or Instruction Book, as applicable, to identify the particular tube. The symbol used in the handbook to identify the tube socket shall appear on the reverse side of the chassis adjacent to the socket. If the available space does not permit such markings of tube type designations and of reference symbols for tubes and tube sockets, a location diagram shall be placed where it is readily visible when viewing the tubes in place or the bottoms of the tube sockets as applicable.

3.38.3.4 *Panels.*—The visible surface adjacent to panel facilities such as controls, indicators, jacks, sockets, and fuse holders shall be marked with a suitable word, phrase, or abbreviation indicating the use or purpose of the part. These markings shall provide good legibility and shall be white. Auxiliary lettering where required, shall be suitably color coded. Continuously variable operating controls shall be provided with markings which will permit the operator to set the control easily and correctly to the predetermined point. Identification markings for these controls, indicators, jacks, sockets and fuse holders shall be approved by the bureau or agency concerned.

3.38.3.5 *Effect on performance.*—Markings shall not adversely affect the leakage path between conductors or any other factor of performance.

3.38.3.6 *Date markings.*—No indication of date of manufacture shall appear on any name plate of the test set.

3.38.4 *Methods of marking other than name plates.*—Methods of marking other than name plates shall be in accordance with the following:

3.38.4.1 When the item or part is a molded glass or plastic product, the markings shall be molded either in raised or depressed characters, or sand blasted.

3.38.4.2 When the item or part is a glazed material, such as porcelain, and has no metal parts, the markings shall be molded, stamped, or marked in ink before the item or part is put through the glazing operation.

3.38.4.3 When the item or part is made from an insulating material in sheet or rod form the marking on it shall be stenciled, stamped, or engraved.

3.38.4.4 When the item or part is soft rubber, or a similar material, the marking preferably shall be impressed or indelibly marked in ink, if the qualities of the items are not impaired. If such a procedure is impracticable, the item shall be supplied wrapped in packages with the marking of the item marked on the outside of the packages.

3.38.4.5 When the item or part is of metal the marking on it shall be stamped with a waterproof ink, etched, cast or engraved. A label as specified in 3.38.4.10 may be used.

3.38.4.6 When the item or part is too small, or its nature prohibits placing a marking thereon, the marking shall appear on the containing package, or a noncorrosive metal tag with the type designation stamped thereon shall be attached to the item or part.

3.38.4.7 When the item or part is a cord, the marking shall be stamped or etched on a metal

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band which shall be securely attached to the cord so that it cannot slip off.

3.38.4.8 When the item or part is supplied in a paper or cloth container, the marking shall be indelibly marked on the container in large conspicuous characters.

3.38.4.9 Wherever practicable, the marking shall be placed in such a location that it will be readily visible.

3.38.4.10 *Labels.*—Labels showing wiring and schematic diagrams, lubricating and operating instructions, safety notices, lists of tools, lists of contents, and similar information shall meet the following requirements:

3.38.4.10.1 *Lettering.*—Gothic type lettering shall be used.

3.38.4.10.2 *Border.*—A border of ¼-inch or more shall be provided on each edge of the label.

3.38.4.10.3 *Legibility.*—Labels shall be legible, and shall be designed to remain so for the service life of the test set on which they are mounted.

3.38.4.10.4 *Materials and processes.*—The material and process used for fabrication of labels shall be such that the finished product shall be satisfactory when mounted on the test set. One of the following materials and processes shall be used:

a. Reverse etched on aluminum plate.

b. Printed on a good grade of white book paper and laminated between two sheets of clear transparent vinylresin which are bonded together to seal against moisture.

c. Lithographed or printed on a suitable plate.

d. Etched on a suitable plate, and filled with permanent contrasting color.

e. Silk screened marking on a suitable plate.

f. *Decalcomanias.*—Those having water-soluble coatings shall not be used.

g. Printed on graphic lamicoid (Mica Insulator Co., New York, N. Y.) or equal.

h. Rubber stamping with waterproof ink.

3.38.4.10.5 *Mounting.*—Labels, except rubber-stamping and decalcomania, shall be securely and permanently mounted using screws or rivets.

3.39 Diagram, schematic and special instruction sheet.—Test sets that are "RESTRICTED" or "UNCLASSIFIED" shall have a schematic diagram and brief operating information affixed to the inside of the cover or to the chassis in a free space to which ready reference can be made at all times during servicing. The backs of removable panels or covers are recommended places for such mountings.

3.39.1 The diagrams shall contain either the actual values of the parts used, such as resistors in ohms and capacitors in microfarads, or a parts list shall be part of the label. The issue of each diagram shall be identified. The diagram shall use electrical graphical symbols in accordance with Standard JAN-STD-15.

3.39.2 The labels shall be in accordance with 3.38.4.10.

3.39.3 The special instruction sheet and schematic diagram shall be sealed in transparent nonburning plastic, or equivalent, for protection.

3.40 Handbook stowage.—Wherever possible, provision shall be made in the operating portion of the test set for the convenient stowage of the Handbook of Maintenance Instructions, Technical Manual, or Instruction Book as applicable.

3.41 Design.—The detailed mechanical and electrical design of the test set shall be accomplished by the contractor subject to the requirements of this specification and the individual test set specification. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance, and permanence of the same. The design, layout, and assembly of the units and their component parts shall be such as to facilitate quantity production. Where drawings or a model are supplied to the contractor for his guidance (unless otherwise stated in the detail specification), this specification and any specification to which it is subsidiary shall govern over the drawings or model. The drawings or model may be followed so far as conformity with this specification and the detail specification exists. Before proceeding with the design, the contractor should consult the bureau or agency concerned concerning discrepancies which may exist in the drawings or model.

3.41.1 The width (the overall width of the chassis, behind the panel including dust cover)

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of all test sets shall be restricted to a maximum of 17 inches so that any piece of standard gear will fit between the rack uprights of a standard 19 inch rack.

3.41.2 *Convenience.*—Each test set and subassembly forming a part of a test set, shall provide easy and ready access to its interior parts, terminals, and wiring, for complete circuit checking and for the removal and replacement of parts. As a general rule it will not be acceptable to displace, or remove wires, cables, parts or assemblies in order to gain access to terminals, soldered connections, mounting screws and the like. Where it is not practicable to avoid such construction, those parts which must be displaced or removed shall be so designed, mounted, and otherwise arranged as to facilitate their displacement or removal when necessary. When in the process of checking or removing a given part it is necessary to displace some other part, the latter wherever practicable, shall be so wired and mounted that it can be moved sufficiently without disconnection from its circuit.

3.41.2.1 Removable containers shall be removable without the necessity of disconnecting any of the leads to parts inside the container.

3.41.2.2 Meters shall be removable from the exterior side of the front panel or subpanel on which mounted, after removal of fastening devices and before removal of electrical connections other than any radio frequency leads connected thereto, except where the test instrument is mounted in a hermetically sealed instrument or combination case.

3.41.2.3 Parts which may require replacement, including those which are identified as maintenance parts in the Parts Catalog Technical Manual or Table of Replaceable Parts of Handbooks of Maintenance Instructions for the test set shall be easily removable. They shall not be mounted or attached by means of rivets or spot welding. Only permanent assemblies (such as chassis, handles, etc.) may be mounted or attached in these ways.

3.41.3 *Ease of adjustment.*—All controls shall be arranged in such a manner and be of such size, construction, and spacing that they are convenient and simple to operate under all service conditions, including those requiring an operator wearing gloves. (See 3.7.) The test set shall be so designed that it can be operated by inexperienced personnel with a minimum of special training.

3.41.4 *Interchangeability.*

3.41.4.1 In order to facilitate the procurement of replacement parts the design shall not be based on the use of parts of special manufacture where suitable units of standard manufacture are available. Similarly, the design shall not be based upon the use of parts provided by only one manufacturer where equivalent parts available from several manufacturers may be employed.

3.41.4.2 In the construction of test equipment, positive action shall be taken to provide for design tolerances sufficient to accommodate various replacement parts such as tubes, instruments, resistors, and capacitors, having the limiting dimensions and mechanical and electrical characteristics required in the specifications for the particular parts involved, without departure from the specified performance.

3.41.4.3 The electrical design of the test set shall be such that operation shall not depend upon special selection of replacement plug-in parts, such as tubes, vibrators, and crystals.

3.41.5 *Life.*—The test set shall be designed to have a reliable operating life of at least 200 hours without requiring servicing. Parts requiring servicing at the end of this interval shall be designated by the contractor. The test set shall be designed to have a minimum operating life of 2000 hours with reasonable servicing and replacement of parts requiring replacement within this interval and the expected life of such parts shall be recommended by the contractor and shall be satisfactory to the bureau or agency concerned.

3.41.6 *Moisture pockets.*—All pockets, wells, traps and the like in which water and condensed moisture can collect when the test set is in normal operating position shall be eliminated or properly drained.

3.41.7 *Lubrication.*—The test set shall be so designed that satisfactory lubrication may be made with standard lubricants. Standard lubricants are those which are in use throughout the Army, Navy, and Air Force.

3.41.8 Two sets of suitable lubrication charts shall be furnished with the test set unless such

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charts are declared unnecessary by the cognizant bureau or agency.

3.42 Safety and protective devices.

3.42.1 Adequate safeguards shall be provided to protect personnel from exposure to high voltages during operation or adjustment. The design and contruction shall be such that the outside cases and frames of all units, that is, all external parts exclusive of antenna or transmission line terminals, shall be at chassis potential when the test set is in operation. Satisfactory operation shall be obtained with the test set standing on a grounded deck. Satisfactory provision shall be made to prevent operating personnel from accidentally coming in contact with potentials in excess of 40 volts. This requirement shall be interpreted to include protection to operating personnel against capacitor discharge. The use of line voltage filters with capacitors not greater than 0.001 microfarad between each side of the line and the chassis is a permissible exception to these requirements. The design of the test set shall be such that the test set will operate safely and in accordance with the applicable specifications, with a ground on either side of the power source line.

3.42.2 When the operation and/or maintenance of a test set employing potentials in excess of 1000 volts requires that these voltages be measured, the test set shall be provided with test points such that all these high voltages may be measured at potential levels less than 1000 volts relative to ground. This may be accomplished through the application of voltage dividers or other techniques such as the use of safety type panel meters and multipliers. Full details shall be given as to the method used in the test set to obtain the voltages at the test points. The above voltages shall be construed as applying to d. c., a. c., or d. c. plus a. c. voltages.

3.42.3 All electrical indicating instruments, with associated multipliers or transformers, except self-contained thermocouple instruments employed in transmitter and rectifier units, shall be protected against damage due to stray radio frequency potentials.

3.42.4 The design of the electrical circuits and controls shall be liberal. Parts shall not be subjected to voltages or currents in excess of the ratings approved by the manufacturer of these parts or the ratings given in the applicable specification, whichever is the lesser. All parts which are likely to carry a heavy overload, due to poor adjustment or tube casualty, for example, shall be proportioned to care for such a load. Where this is impracticable, fuses or other suitable devices shall be included to protect the parts.

3.42.5 Where voltages in excess of 125 volts occur within the test set case, there shall be placed in a suitable location, which may be inside the case, a warning notice stating approximately the highest voltage which may be encountered.

3.42.6 Suitable provisions shall be made to avoid fire hazard. In addition to nonburning materials and fireproof insulation, adequate protective devices, such as fuses or circuit breakers, shall be incorporated in the primary circuits of the test set to eliminate the development of extreme temperature resulting from excessive current due to defects in the test set or overload. Both sides of the primary line shall be protected or fused.

3.42.7 *Fuses and fuse holders.*

3.42.7.1 All fuses shall be mounted in finger-operated extractor fuse posts.

3.42.7.2 The insulating cap of each fuse post shall have a hole permitting the insertion of a test prod for contact with the metallic frame holding the load end of the fuse. The test prod concerned with this requirement is shown on Drawing SC–C–6878 and SC–D–6883.

3.42.7.3 All fuses shall be readily accessible from the front panel. There shall be one spare loaded fuse post adjacent to each active fuse normally used.

3.42.7.4 All fuses shall be of the 3AG or 4AG commercial type.

3.42.7.5 The rating of each fuse as indicated on the fuse shall be marked adjacent to each fuse post. In addition, "SPARE" shall be marked adjacent to each spare fuse post.

3.42.8 *Circuit breakers.*—If circuit breakers are used the restoring or switching handle shall be readily accessible from the front panel.

3.43 Interference.—Test sets shall be protected from external interference created by the operation of other nearby equipments. Test sets shall not generate electrical interference

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which will adversely affect the operation of other nearby equipments nor feed back interference or unwanted signals on external lines.

3.43.1 *Radio frequency interference (noise).*—The test set shall have maximum practicable protection from the effects of electrical and radio frequency noise and interference signals, and have provisions against the generation of undesired signals which might adversely affect other equipments operated in the same vicinity.

3.43.1.1 *Interference noise generated by the test set.*

3.43.1.1.1 *Conducted interference.* — The conducted radio frequency voltages (interference) produced by the operation of any item of the test set, on wiring connected to or associated with the test set and as measured across any two terminals and between any terminal and ground or frame of the test set shall not exceed 5.0 microvolts over a frequency range of 0.015 megacycle to 400 megacycles.

3.43.1.1.2 *Radio interference (noise) field.*—The undesired radio frequency interference field produced by the operation of any item of the test set shall not exceed 5.0 microvolts per meter over the frequency range of 0.015 megacycles to 400 megacycles. The sensing antenna of the measuring equipment shall be mounted in various positions 1 foot from any part of the test set, its antenna or antennae and interconnecting cable assemblies being tested in order to obtain both vertical and horizontal components of the interference field; and at no point shall the interference field exceed the above limits. This shall not preclude the use of heterodyne type oscillators and frequency meters, when required in the detailed specification, but undesired interference fields shall be kept to a minimum.

3.43.1.2 *Susceptibility to conducted noise generated by other equipments.*—A signal of any value up to 1000 microvolts injected into any external circuit or lead, except the antenna, shall produce no degradation in electrical performance of visual or other circuits or shall produce no change in the output. The signal shall be any frequency throughout the range of the lowest frequency above audio used in the test set to 400 megacycles and shall be (a) unmodulated and (b) modulated 30 percent with either 400 or 1000 cycles. In these tests, squelch and noise limiter circuits shall be disabled.

3.43.1.3 These tests can be conducted satisfactorily by use of the following equipment:

a. Standard receiver equipment employed as transfer devices in conjunction with a signal generator and a calibrated output meter for measurements over the frequency range of 0.015 megacycle to 0.150 megacycle, until there becomes available standard radio interference and field intensity measuring equipment covering this frequency range.

b. Ferris Radio Noise and Field Strength Meter, Model 32-A; over the frequency range of 0.150 megacycle to 15 megacycles.

c. Army-Navy type TS-587/U Radio Interference and Field Intensity Measuring Equipment over the frequency range of 15 megacycles to 400 megacycles.

d. Any measuring equipment satisfactory to the bureau or agency concerned.

Tests shall be made using an arrangement of equipment, interconnecting cable assemblies, and supporting structure which approximate service conditions.

3.43.2 *Shielding.*—The number of mechanical discontinuities in the case, such as covers, inspection plates, front plates, joints, indicating meter, holes, etc., shall be kept to a minimum. All necessary mechanical discontinuities in the case shall be electrically continuous across the interface of the discontinuity so as to provide a low impedance current path throughout the frequency range covered by the individual test-set specification.

3.44 Standby heater. — Each externally powered test set, except battery operated, shall be provided with a standby heater energized when the power switch is not in the "ON" position to raise the temperature of the set when outside of the transit case, at least 6° C. above ambient temperature.

3.45 Power switch.—Application of power to each externally powered test set shall be controlled by a power ON-OFF switch located on the front panel. An indicator lamp shall be provided to indicate application of power to the test set, except for dry battery operated, low power equipment.

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3.46 Service conditions (atmospheric).—The test set shall be so designed and constructed that no fixed part or assembly shall become loose, no moving or movable part or assembly shall become undesirably free or sluggish in operation, and no degradation shall be caused in the performance below that specified in the individual specification for the particular test set when subjected to the following conditions (see 4.4):

3.46.1 *Test set operating.*—After a 20 minute warm-up period, the test set shall operate under any possible combination of the following ambient conditions.

3.46.1.1 *Temperature.*

3.46.1.1.1 Continuous operation over the range of −54° C. (−65° F.) to +65° C. (+150° F.).

3.46.1.1.2 Operation at 72° C. (161.6° F.) for a continuous period of 20 minutes. It is not expected that the specified accuracies will be obtained during operation at 72° C. (161.6° F.).

3.46.1.2 *Humidity.*—Operation shall be possible at relative humidities ranging up to 95 percent and above for both continuous and intermittent periods, including conditions wherein condensation takes place in and on the test set in the form of both water and frost.

3.46.1.3 *Altitude.*—Operation at barometric pressures ranging down to 20.6 inches of mercury (approximating an altitude of 10,000 feet).

3.46.2 *Test set nonoperating.*—The following conditions and any possible combination thereof are those to which the test set may be subjected while inoperative and stand-by heater not connected. The test set shall sustain no degradation in performance below that specified in the individual specification for the particular test set after retuning to the operating range service conditions. (See 3.46.1 and 4.4.)

3.46.2.1 *Temperature.*—Ranging from −65° C. (−85° F.) to 85° C. (185° F.) for extended periods.

3.46.2.2 *Humidity.*—Ranging up to 100 percent relative for extended periods.

3.46.2.3 *Altitude.*—Barometric pressures ranging down to 3.4 inches of mercury approximating an altitude of 50,000 feet.

3.46.2.4 *Fungus.*—Subjected for extended periods to fungus and fungus supporting conditions.

3.47 Service conditions (mechanical).—The test set shall be so designed and constructed that no fixed part shall become loose, no movable part or control shall be shifted in setting, position, or adjustment, and no permanent degradation shall be caused in performance below that specified in the individual specification for the particular test set when subjected to the following conditions (see 4.4):

3.47.1 *Test set operating.*

3.47.1.1 *Shock.*—The test set shall not suffer damage, or subsequently fail to give the performance required by the individual specification for the particular test set, when subjected to shock test specified in 4.4.6.

3.47.1.2 *Vibration.*—The test set shall not suffer damage or fail to give the performance required by the individual specification for the particular test set, when subjected to vibration in accordance with the requirements of 4.4.5.

3.47.2 *Test set, nonoperating.*

3.47.2.1 *Mounting.*—No impaired performance or damage to the test set shall result from storage in any position.

3.47.2.2 *Shock.*—No impaired performance or damage to the test set shall result from shocks as specified in 4.4.6.

3.47.2.3 *Vibration.*—No impaired performance or damage to the test set shall result from vibrations as specified in 4.4.5.

3.48 Service conditions (electrical).—The test set shall be so designed and constructed that no degradation in performance below the requirements of the individual specification for the particular test set shall result when the test set is operated from the primary power and under the conditions herein specified.

3.48.1 *Internally powered test sets.*—Internally powered test sets shall be designed for the use of self-contained batteries which are in accordance with Specification JAN-B-18. (See 3.52.)

3.48.2 *Externally powered test sets.*

3.48.2.1 *Alternating-current powered.*

3.48.2.1.1 The test set shall operate from 115 volts, alternating current, ±10 percent, over the frequency range of 50 to 1,000 cycles per second. The alternating current supply voltage may have a form factor varying over the

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range, 1.25 to 1.03 and may have a crest factor varying over the range 1.68 to 1.16. Form factor is defined as the ratio of the effective voltage to the half period average voltage, and has the value of 1.11 for a true sine wave. Crest factor is defined as the ratio of the crest voltage to the effective voltage, and has a value of 1.414 for a true sine wave. Manually operated devices such as tapped transformers, variable voltage transformers (variacs or equivalent) and the like shall not be employed unless specifically authorized in the individual test set specification.

3.48.2.1.2 Neither side of the supply voltage shall be directly connected to the chassis.

3.48.2.1.3 *Resonant points.*—Particular attention shall be paid to elimination of circuits resonant to frequencies within the specified range of the power source and at beat frequencies between the power frequency and other frequencies present in the particular test set.

3.48.2.2 *Direct-current powered.*—The test set shall operate from whichever one or more of the following power supplies specified in the individual test set specification.

Power supply voltage d. c.
26 volts ± 6
13.5 volts ± 3
6.75 volts ± 1.5

3.49 Composition of a complete test set.—A complete test set shall comprise the following, unless otherwise specified in the individual test set specification:

a. Test instrument proper (major assembly), including any tubes, piezo-electric crystals, or rectifier crystals needed for operation.

b. Transit case (unless the combination case construction is used.) (See 3.2.1.)

c. Set of accessories (minor assemblies).

3.50 Degree of enclosure.

3.50.1 *Transit cases and combination cases.*—Transit cases and combination cases shall be watertight as defined in Specification 17E11, unless otherwise specified in the individual test set specification.

3.50.2 *Instrument cases.*—Instrument cases shall be dripproof in accordance with Specification 17E11.

3.51 Case.—The individual specification for the particular test set or the covering contract or order will specify whether a transit case or a combination case is to be supplied by the contractor.

3.51.1 *Transit case.* (*See 3.2.1.*)

3.51.1.1 *Weight.*—The weight of the case when loaded shall be kept to a minimum.

3.51.1.2 *Weight and torque distribution.*—The weight and torque distribution per handle shall be as even as practicable when the case is loaded.

3.51.1.3 *Volume.*—The total volume occupied by the case shall be kept to a minimum.

3.51.1.4 *Material.*—The case shall be constructed of aluminum or a three or more ply material, phenolic-resin-bonded, having the outer plies of aluminum and inner ply or plies of a suitable material. Where the capacity of the case is less than 1 cubic foot, it may be constructed of aluminum.

3.51.1.5 *Cover.*—The case shall have a nondetachable cover secured to the case by hinges, chains or other suitable means.

3.51.1.6 *Handles.*—Hinged metal handles shall be provided, having a clearance from the case sufficient to pass a block 1⅞ by 4⅜ inches in cross section and having the edges rounded to a $^{15}/_{16}$ inch radius. Handles shall be of the casket type if they are on the side of the case. The grip portion of the handle shall be of a non-metallic material and shaped to fit the hand comfortably.

3.51.1.7 *Latches.*—The case shall be provided with trunk-type pull-down latches, i. e., hooks in the case and latches in the cover.

3.51.1.8 *Corners.*—Cases other than all-metal types shall be provided with metal reinforced corners. All corners shall be rounded with at least ¼ inch radius.

3.51.1.9 *Attachment of name plate.*—The name plate shall be firmly attached to the inside of the cover of the case.

3.51.1.10 *Accessory storage.*—The case shall have provision in its cover or in a separate compartment for the stowage of all accessories.

3.51.2 *Combination case.* (*See 3.2.3.*)

3.51.2.1 *Weight.*—The weight of the case when loaded shall be kept to a minimum.

3.51.2.2 *Material.*—The case shall be constructed either of aluminum, or of a three or more ply material, phenolic-resin-bonded, hav-

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ing the outer plies of aluminum and inner ply or plies of a suitable material. Where the capacity of the case is less than 1 cubic foot, it may be constructed of aluminum. The front panel shall be of aluminum.

3.51.2.3 *Cover.*—The cover shall be attached by means of separable hinges.

3.51.2.4 *Handle.*—One hinged metal handle shall be provided on top of the case. The handle shall have a clearance from the case sufficient to pass a block 1⅞ by 4⅜ inches in cross section and having the edges rounded to a 15/16 inch radius. The grip portion of the handle shall be of a nonmetallic material and shaped to fit the hand comfortably.

3.51.2.5 *Panel protection.*—Controls and indicators shall be protected by recessing the panel or by providing two rigid metal handle guards.

3.51.2.6 *Latches.*—The case shall be provided with trunk type, or spring loaded pull down latches.

3.51.2.7 *Case fasteners.* — The fasteners which attach the case to the chassis shall be of the captive type.

3.51.2.8 *Corners.*—Cases other than all-metal types shall be provided with metal reinforced corners. All corners shall be rounded with at least ¼ inch radius.

3.51.2.9 *Ventilation.*—Any ventilation required shall be through louvers. The ventilating apertures shall be protected with 20-mesh or finer screen. Any exhaust air resulting from forced ventilation shall be directed so that it will not inconvenience the operator.

3.51.2.10 *Attachment of name plate.*—The name plate shall be firmly attached to the front panel.

3.51.2.11 *External connections and controls.*—All external electrical connections and all operating controls shall be on the instrument panel.

3.51.2.12 *Accessory stowage.*—The case shall have provision in its cover or in a separate compartment for the stowage of accessories and operating spares.

3.51.3 *Instrument case (see 3.2.2).*—An instrument case shall be used when a transit case is provided.

3.51.3.1 *Materials.*—The case and front panel shall be constructed of aluminum.

3.51.3.2 *Handles.* — If carrying handles, other than the panel guards specified in 3.51.3.3, are required by the individual test set specification, they shall be of metal and hinged, and shall have a clearance from the case sufficient to pass a block 1⅞ by 4⅜ inches in cross section, and with the edges rounded to a 15/16 inch radius. The grip portion of the handle shall be of nonmetallic material and shaped to fit the hand comfortably.

3.51.3.3 *Panel protection.*—Controls and indicators shall be protected by recessing the panel or by two rigid metal handle guards.

3.51.3.4 *Case fasteners.*—The fasteners which attach the case to the chassis shall be of the captive type.

3.51.3.5 *Feet.*—The case shall be provided with two sets of four feet each, or dimples or molded projections of the case, one set on the bottom and one set on the rear.

3.51.3.6 *Ventilation.*—Any ventilation required shall be through louvers. The ventilating apertures shall be protected with 20-mesh or finer screen. Any exhaust air resulting from forced ventilation shall be directed so that it will not inconvenience the operator.

3.51.3.7 *Attachment of name plate.*—The name plate of the instrument shall be firmly attached to the instrument panel.

3.51.3.8 *External connections.*—All external electrical connections and all operating controls shall be on the instrument panel unless otherwise authorized by the bureau or agency concerned.

3.52 **Batteries.**

3.52.1 No batteries shall be supplied with the test set.

3.52.2 Battery-powered test sets shall be provided with a time switch or an automatic cut-off switch which opens the circuit when the lid of the combination case is closed.

3.52.3 The battery compartment shall be so constructed that the batteries will be rigidly mounted and adequate room shall be provided so that deteriorated batteries may be removed readily without damage to the test set. Adequate protection shall be provided so that no injury to the test set will result from emitted chemicals or fumes from the batteries either in in a normal or deteriorated state.

3.52.4 The battery compartment in the test

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set shall be marked to indicate connections, polarity, voltage, and type of batteries required.

3.52.5 Battery-powered test sets shall be labeled externally as follows:

WARNING

REMOVE BATTERIES BEFORE SHIPMENT OR INACTIVE STOWAGE OF 30 DAYS OR OVER.

3.53 Mounting provisions. — Provisions shall be made for mounting the equipment to a deck or bulkhead (floor or wall) when specified in the individual equipment specification.

3.54 Workmanship.—

3.54.1 *General.*—The test set, including all parts and accessories, shall be manufactured and finished in a thoroughly workmanlike manner. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, painting, riveting machine-screw asemblage, welding, and brazing.

3.54.2 *Dimensions.*—All dimensions and tolerances not specified shall be consistent with the best shop practices. Where dimensions and tolerances affect the interchangeability, operation, or performance of the test set, they shall be held or limited accordingly.

3.54.3 *Fabrication.*—All machining, drilling, and forming shall be performed by the use of accurate templates, jigs or gages. Boxes, cases, shields, and compartment walls shall be made by casting, drawing, or bending and welding, except where ease of servicing of the test set requires that a removable panel construction be used or where the applied stresses dictate the use of a strong aluminum alloy which does not provide a good weld. For such parts riveting or bolting may be used.

3.54.4 *Screw assemblies.* — All assembly screws shall be tight. The term "tight", in this instance means that the screw cannot be appreciably tightened further without danger of injury to the screw or threads.

3.54.5 *Riveting.*—Riveting operations shall be carefully performed in order to insure that the rivets are tight and satisfactorily headed.

3.54.6 *Gears.*—All gear assemblies shall be properly alined and meshed, and shall be operated without interference, tight spots, loose spots, excessive backlash, or other irregularities.

3.54.7 *Clean-up.*—After final assembly all units shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips and other foreign material.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Classification of tests.—The methods of sampling, inspection and tests conducted upon test sets shall be classified as follows:

4.1.1 *Preproduction tests.*—Preproduction tests are made on a sample(s) representative of the production test sets, to determine that the production test sets will meet the requirements of this specification and the individual test set specification.

4.1.2 *Inspection tests.*—Inspection tests are made on production test sets submitted for acceptance under the contract to determine conformance with the requirements of this specification and the individual test set specification and to determine that the test sets are equivalent in performance and construction to the approved preproduction sample. Any production prior to approval of the preproduction sample shall be at the contractor's risk.

4.2 Preproduction tests.—Inspection and test shall be made to determine whether the contractor's design conforms to the requirements of this specification and of the individual specification for the particular test set. Preproduction tests shall consist of all tests specified in 4.4 and any additional tests specified in the individual test set specification.

4.2.1 *Design test sets and accompanying data.*

4.2.1.1 *Experimental contracts.*—For contracts covering the design, development, and furnishing of a specified small quantity of experimental test sets without associated large scale production on the same contract the contractor shall supply complete test sets in the quantity as specified in the contract or order. (See 6.1.) Each experimental test set shall include and have packaged with it the following:

4.2.1.1.1 One copy of operating instructions.

4.2.1.1.2 One copy of an over-all functional schematic circuit diagram reduced to its simplest form and showing the circuits of the complete test set without reference to the actual physical location of the circuit elements in the

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different major and minor assemblies and external detail parts, and omitting references to interconnection through plugs, recptacles (sockets), and the like. Each experimental test set will be subjected to such of the preproduction test procedures specified in 4.4 as are required by the bureau or agency concerned.

4.2.1.2 *Production contracts.*—For production contracts the following contractor's design sample equipment and accompanying data shall be submitted for test and approval of design prior to large scale production by the contractor. Approval of design shall also be subject to specific approval of the sample parts and materials and the design data required as specified in 4.2.2 and 4.2.3, respectively.

4.2.1.2.1 One or more complete test sets as specified in the contract or order shall be submitted. The test sets shall be of the preproduction design category unless the bureau or agency concerned authorized delay in submission of the design sample test set(s) until a small number of production test sets have been manufactured. Any such authorization for first articles of production to be submitted as design sample test sets does not relieve the contractor of the responsibility to furnish the samples of parts and materials and design data specified in 4.2.2 and 4.2.3 in advance of the submission of such first article test sets as design sample test sets. Upon completion of the preproducttion tests, in accordance with the requirements of 4.4, at least one of the complete test sets will be returned to the contractor, for use until a production test set has been established by the bureau or agency concerned as a standard model for the continuance of production. Each sample test set may be conditioned by the contractor for submission as a production set.

4.2.1.2.2 One additional complete set of tubes shall be furnished.

4.2.1.2.3 One set of operating instructions shall be supplied with each sample test set.

4.2.1.2.4 There shall be packed with each sample test set required, one copy of an overall functional schematic circuit diagram reduced to its simplest form and showing the circuits of the complete test set without reference to the actual physical location of the circuit elements in the different major and minor assemblies and external parts, and omitting reference to interconnection through plugs, receptables (sockets), etc.

4.2.2 *Sample parts and materials.*—The sample parts and materials may be tested to destruction by the testing activity and may not be returned to the contractor. The following shall be furnished:

4.2.2.1 Standard parts and materials shall be furnished for which qualification is required by subsidiary specifications but for which such qualification has not already been certified, and in the quantities required by the applicable specifications.

4.2.2.2 Nonstandard parts and materials shall be furnished of identities and quantities as designated by the bureau or agency concerned upon examination of the stock list specified in 4.2.2.4. Not more than twelve of each type of part will be required.

4.2.2.3 With the submission of nonstandard parts and materials, the contractor shall furnish simultaneously, complete test data to show that they will satisfactorily meet the requirements of this specification, the individual test set specification, and any other pertinent subsidiary specification.

4.2.2.4 *Preliminary stock list.*—Three copies of a preliminary stock list of parts including all electrical parts, identified by Joint Army-Navy (JAN) type designation or manufacturer's name, part number, and rating if not a JAN type shall be furnished. Where practicable, the list shall be sectionalized according to individual major and minor assemblies.

4.2.2.4.1 The preliminary stock list together with the overall functional schematic circuit diagram and the complete schematic wiring diagram shall be furnished directly to the bureau or agency concerned in advance of the submission of the design data, in order that the bureau or agency concerned may review the list of parts to be used in the test set and desig nate parts which are to be submitted for preproduction test.

4.2.2.5 *Identifying data.*—All sample parts and materials submitted shall be accompanied by accurate identifying information, including the following:

a. Government order number (file number) or contract number.

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b. Prime contractor's name and part number.

c. Manufacturer's name and part number.

d. An accurate description of the part or material, including the following:

(1) Number of the standard specification (JAN, ARMY, ANA, etc.) under which part or material was manufactured.

(2) Special features of construction.

(3) Materials used (type, grade, etc.).

(4) Type of insulation, dielectric, etc.

(5) Type of finish or covering.

(6) Details of spring materials, contacts, etc.

(7) Details of tropicalization process.

(e) Data concerning previous approval, if the part has been previously submitted. Such approval and information shall not release the contractor from complying with the requirements of this specification as to submission of sample parts and materials.

4.2.3 *Design data.*—The following information shall be furnished except that on an experimental contract only, items a, b, and c may be omitted:

a. Three copies of a complete cabling diagram of the complete test set.

b. Three copies of outline dimensional sketches of all major and minor assemblies and of any external detail parts, showing projections.

c. Three copies of specifications for all tubes which are not covered in Specification JAN–1.

d. Three additional copies of the operating instructions specified in 4.2.1.1.1 or 4.2.1.2.3, as applicable.

e. Prior to submission of the sample, three copies of a report by the contractor of his prior tests on the test sets, including the preproduction test as specified in the individual test set specification with an analysis of all failure which occurred and suggestions for improvements in design which might be incorporated in later productions.

f. Seven additional copies of the overall functional schematic circuit diagram specified in 4.2.1.1.2 or 4.2.1.2.4, as applicable.

g. Three copies of complete circuit and cabling diagrams, showing all circuits used in the test set. The diagram shall be in accordance with examples shown on Drawings SC–D–15800, SC–D–15801, SC–D–20095, SC–D–20096, and SC–D–20097.

h. Three copies of a practical wiring diagram of each major assembly or of each constructional unit of the assembly, whichever is practicable, showing the physical location and connections of detail parts and subassemblies with reference symbols and terminal numbers indicated.

4.2.4 *Presubmission testing.*—No item, part, or complete test set shall be submitted to the Government for approval until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.3 Inspection tests.—Inspection tests shall be as specified in the individual test set specification.

4.4 Test methods and procedures.—The test methods and procedures are intended to simulate actual conditions that may be encountered in the storage, shipment by air, land, or water, and nonflight operation of electronic test equipment. Any transit case shall be separated from the instrument proper, and the cover of any combination case shall be separated from the case proper. Where the methods and procedures for conducting tests are not stated in this specification or in the individual test set specification, they shall be prepared by the contractor and sent to the bureau or agency concerned for approval. Any additional tests considered necessary, by the bureau or agency concerned, to determine conformance with the requirements of the individual test set specification shall be conducted.

4.4.1. *Preconditioning.* — Preconditioning shall be accomplished in a temperature altitude chamber with the test set nonoperating. To check temperature stabilization, place a thermocouple in good thermal contact on the largest

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internal mass centrally located, or in any other location deemed necessary by the Government inspector. The temperature cycling and altitude (barometric pressure) cycling may be accomplished separately rather than in one procedure.

4.4.1.1 Raise the temperature to 85° C. (185° F.) and maintain that temperature until temperature stabilization has been reached.

4.4.1.2 Lower the temperature to −65° C. (−85° F.) and the barometric pressure to 3.4 inches of mercury. These two changes may be accomplished in sequence or simultaneously. Maintain those conditions until temperature stabilization has been reached

4.4.1.3 The pressure shall be raised to 30 inches of mercury and the temperature to ambient of 15° C., and maintain until stabilization has been reached. These two changes may be accomplished in sequence or simultaneously.

4.4.1.4 The procedures specified in 4.4.1.1, 4.4.1.2, and 4.4.1.3 shall be repeated for three additional cycles.

4.4.2 *Humidity test.*—The test set to be tested shall be subjected to the following:

a. Dry the test set for 24 hours at 130° --5° F.

b. Measure performance of the test set for compliance with the specifications and make any necessary adjustments to enable the test set to meet the specification.

c. Subject unsealed test sets to five 48-hour cycles as shown on Drawing SC–D–15914, and sealed test sets to 15 of these cycles. Remove sealed test set from its sealing case for the last two cycles.

d. Make test measurements during the periods specified on the drawing, while the test set is exposed to the humidity conditions shown thereon. Make the measurement not more than 5 minutes after power is applied to the test set and complete them as rapidly as possible.

4.4.3 *Fungi test.*

4.4.3.1 Immediately after the test set has been subjected to the humidity test it shall be removed from its protective cabinet, unless hermetically sealed therein, or when cover or cabinet forms an integral part of the mechanism, and the circuit elements exposed and all parts placed in an incubator in which conditions are maintained at 30° C. ±5° C. (86° F ±9.0° F.), 90 to 100 percent relative humidity with some condensation and at atmospheric pressure. The air should be relatively stagnant although a little ventilation should be permitted to occur by convection and leaks through the cabinet. A ventilating fan should not be used.

4.4.3.2 The parts shall be inoculated with a mixture of fungi spores which include the following species: penicillium luteum, aspergillus niger, aspergillus flavus, trichoderma lignorum T–1 U. S. D. A. (see 6.2). Spores shall be mixed with distilled water and approximately ½ ounce atomized (using de Vilbiss type atomizer or equal) into the incubator above the parts but not directly on them. This mixture shall be made up fresh for each test inoculation.

NOTE: Care should be taken not to inhale this mixture, and to wash the hands after handling.

4.4.3.3 The inoculated test set shall be incubated for a period of 28 days. At the end of that time the parts shall be visibly examined for evidence of fungus growth. Any visible growth will be cause for rejection.

4.4.4 *Temperature and altitude tests.*—The following tests shall be conducted with the test set in a temperature-altitude chamber in the order specified. Suitable provisions shall be made to pass the required electrical connections through the walls of the chamber. Test sets requiring internal batteries may use an external power source for that part of the test specified in 4.4.4.6, 4.4.4.7, and 4.4.4.8 if suitable low temperature batteries are not available; operation at all other temperatures shall be made using the internal batteries. To check temperature stabilization, place a thermocouple in good thermal contact on the largest internal mass centrally located or in any other location deemed necessary by the Government inspector.

4.4.4.1 With the test set stabilized at an ambient temperature of 25° C. (77° F.), the test set shall be operated for a 20-minute warmup period, to determine whether the performance requirements of the specification are maintained.

4.4.4.2 With the test set still operating, the ambient temperature shall be lowered to 0° C. (32° F.) and sufficient time for temperature stabilization shall be allowed.

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4.4.4.3 It shall be determined whether the performance requirements of the individual test set specification are maintained and all controls and switches can be readily manipulated.

4.4.4.4 With the test set still operating, the ambient temperature shall be lowered to −20° C. (−4° F.) and sufficient time for temperature stabilization shall be allowed.

4.4.4.5 It shall be determined whether the performance required in the individual test set specification is maintained and all controls and switches can be readily manipulated.

4.4.4.6 With the test set not operating the ambient temperature shall be lowered to −40° C. (−40° F.) and sufficient time for temperature stabilization shall be allowed.

4.4.4.7 The test set shall be turned on for sufficient time to determine whether the test set will function satisfactorily, then the test set shall be turned off to allow temperature stabilization. This procedure shall be repeated three times.

4.4.4.8 The test set shall be operated for the specific warmup period, and immediately determine whether the performance requirements of the individual test set specification are maintained and all controls and switches can be readily manipulated.

4.4.4.9 With the test set still operating, the ambient temperature shall be raised to 25° C. (77° F.) and temperature shall be held at ambient until stabilization has been reached.

4.4.4.10 With the test set still operating, the barometric pressure shall be lowered to 20.6 inches of mercury. After a 15-minute period it shall be determined whether the performance requirements of the individual test set specification are maintained.

4.4.4.11 The pressure shall be raised to ambient atmospheric and the test set shall be operated to determine whether the performance requirements of the individual test set specification are maintained.

4.4.4.12 With the test set still operating, the ambient temperature shall be raised to 55° C. (131° F.) and sufficient time for temperature stabilization shall be allowed.

4.4.4.13 The test set shall be operated for 30 minutes, and it shall be determined whether the performance requirements of the individual test set specification are maintained.

4.4.4.14 With the test set not operating, the ambient temperature shall be raised to 71° C. (159.8° F.) and sufficient time for temperature stabilization shall be allowed.

4.4.4.15 The test set shall be turned on for sufficient time to determine whether the test set will function satisfactorily. Then the test set shall be turned off to allow temperature stabilization. This procedure shall be repeated three times.

4.4.4.16 The test set shall be operated for 20 minutes.

4.4.4.17 With the test set not operating, the ambient temperature shall be raised to 85° C. (185° F.). The temperature shall be held at 85° C. (185° F.) for 30 minutes after temperature stabilization.

4.4.4.18 The ambient temperature shall be lowered to 25° C. (77° F.) and held for 30 minutes after temperature stabilization. The test set shall be inspected for degradation.

4.4.4.19 The test set shall be turned on for sufficient time to determine whether the test set will function satisfactorily. Then the test set shall be turned off to allow temperature stabilization. This procedure shall be repeated three times.

4.4.4.20 The test set shall be operated to determine whether the performance requirements in the individual test set specification are maintained.

4.4.5 *Vibration test.*—The test set installed in its transit or combination case shall be mounted on the vibration table. If internal batteries are required they shall be placed in the case and fastened in a normal manner required for operation. During the vibration the test set shall be energized and monitored.

4.4.5.1 *Variable frequency vibration.*

4.4.5.1.1 *Test 1.*—The test set shall be vibrated with the frequency varying between 10 and 33 cycles per second at a total excursion of 0.06±0.006 inch, the frequency varying uniformly from 10 to 33 cycles per second and returning to 10 cycles per second in approximately ½ minute.

4.4.5.1.2 *Test 2.*—The test set shall be vibrated with the frequency varying between 10 and 55 cycles per second at a total excursion of 0.03±0.006 inch, the frequency varying uniformly from 10 to 55 cycles per second and

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returning to 10 cycles per second in approximately 1 minute.

4.4.5.1.3 Application of the vibration specified in 4.4.5.1.1 and 4.4.5.1.2 shall be in each of the following directions:

a. Horizontally, parallel to the major horizontal axis of the test set, for a period of 15 minutes for each frequency range.

b. Horizontally, at right angles to the major horizontal axis, for a period of 15 minutes for each frequency range.

c. Vertically, for a period of 15 minutes for each frequency range.

4.4.5.2 *Fixed frequency vibration.*—The test set shall then be vibrated for a period of 3 minutes at each of the four most severe resonant frequencies, at the same amplitude as specified in 4.4.5.1, and in the plane observed to give the most severe reaction.

4.4.5.3 The test set shall be inspected to ascertain that no fixed parts have become loose or damaged. The test set shall be operated to demonstrate that the performance required in the individual test set specification is maintained.

4.4.6 *Shock test.*—Unless otherwise specified in the individual specification for a particular test set the test set shall be tested in accordance with the following procedure:

4.4.6.1 *Shock test machines.*—The shock testing machine shall be as shown on Drawing 10-T-2145 as illustrated by figure 2. The test set shall be mounted in such a way as to approximate the actual rigidity encountered in service in the utilization of the particular test set. Portable test sets shall be tested outside the transit case, secured rigidly to the mounting plate of the shock machine except when designed to operate on shockmounts. If a transit case is provided the test set shall also be tested de-energized in the transit case.

4.4.6.2 A total of nine blows shall be applied. Three blows shall be applied parallel to each of the three principal axes of the test set being tested. The height of hammer drop for the back blow and side blow shall be 1 foot, 2 feet, and 3 feet. For the top blow, the height of hammer drop shall be 2 feet, 3 feet, and 4 feet.

4.4.6.3 After each blow of the shock test, the test set shall continue to perform its principal functions without any attention to or replacement of parts. Adjustment will be permitted to external controls only. Momentary failure to operate during shock is permissible providing the test set resumes normal operation after each blow. Minor damage or distortion will be permitted, provided such damage or distortion cannot in any way impair the ability of the test set to perform its principal functions.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be as specified in the individual test set specification.

6. NOTES

6.1 Since this specification is general in scope and covers only the general requirements for the design and manufacture of test equipment used in testing of electronic equipment, the details of performance of the test set under the conditions specified herein and the ordering information must be specified elsewhere. Attention of design engineers is invited to the items listed below which should be covered in the individual test set specification or in the contract or order:

a. Detail performance requirements for the particular test set.

b. Uses of wood other than those listed in 3.1.5.1.3.

c. Requirements for piezo-electric crystal units. (See 3.9.1.)

d. The operational characteristics of any relays needed in the test set and not covered in 3.19.

e. The power supply for direct-current powered test sets. (See 3.48.2.2.)

f. Whether a transit case or a combination case is to be supplied. (See 3.51.)

g. Additional preproduction tests. (See 4.2.)

h. Quantity of test sets required under an experimental contract. (See 4.2.1.1.)

i. Quantity of sample test sets required under a production contract. (See 4.2.1.2.1.)

j. Inspection tests. (See 4.3.)

k. Preparation for delivery. (See 5.1.)

6.2 The fungi referred to in 4.4.3.2 may be obtained from the American Type Culture Collection, Georgetown University School of Medi-

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cine, 3900 Reservoir Road, Washington, D. C. The cultures are specified as follows:

	American Type Culture Collection, Georgetown
Aspergillus Niger	9642
Aspergillus Flavus	9643
Penicillum Luteum	9644
Trichoderma T-1 U. S. D. A	9645

6.3 Salicylanilide is available commercially as Shirlan Extra. Shirlan Extra and Hexalin are manufactured by E. I. Dupont DeNemours and Co., Inc., Wilmington, Delaware. The formula given in 3.36.5.1 is available as follows:

CL-21585	Glidden Co., Chicago, Ill.
SN-37	Insl-X Co., Brooklyn, N. Y.
61-AH8	Maas and Waldstein Co., Newark, N. J.
10704	Vita-Var Corp., Newark, N. J.

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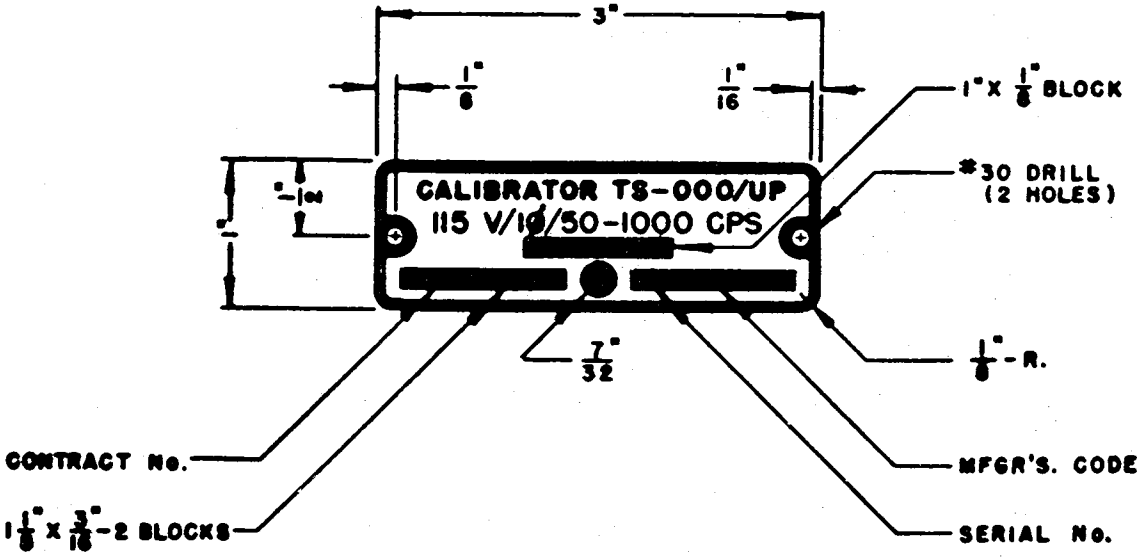
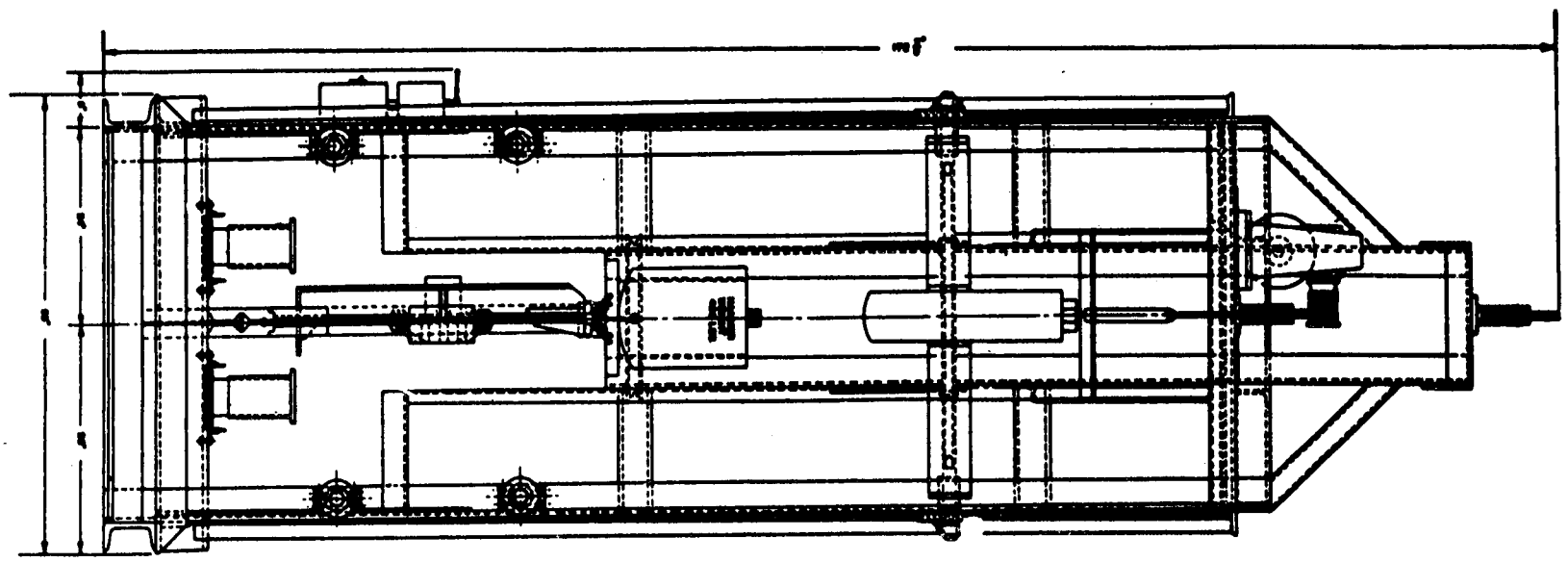


FIGURE 1. *Nomeplate.*

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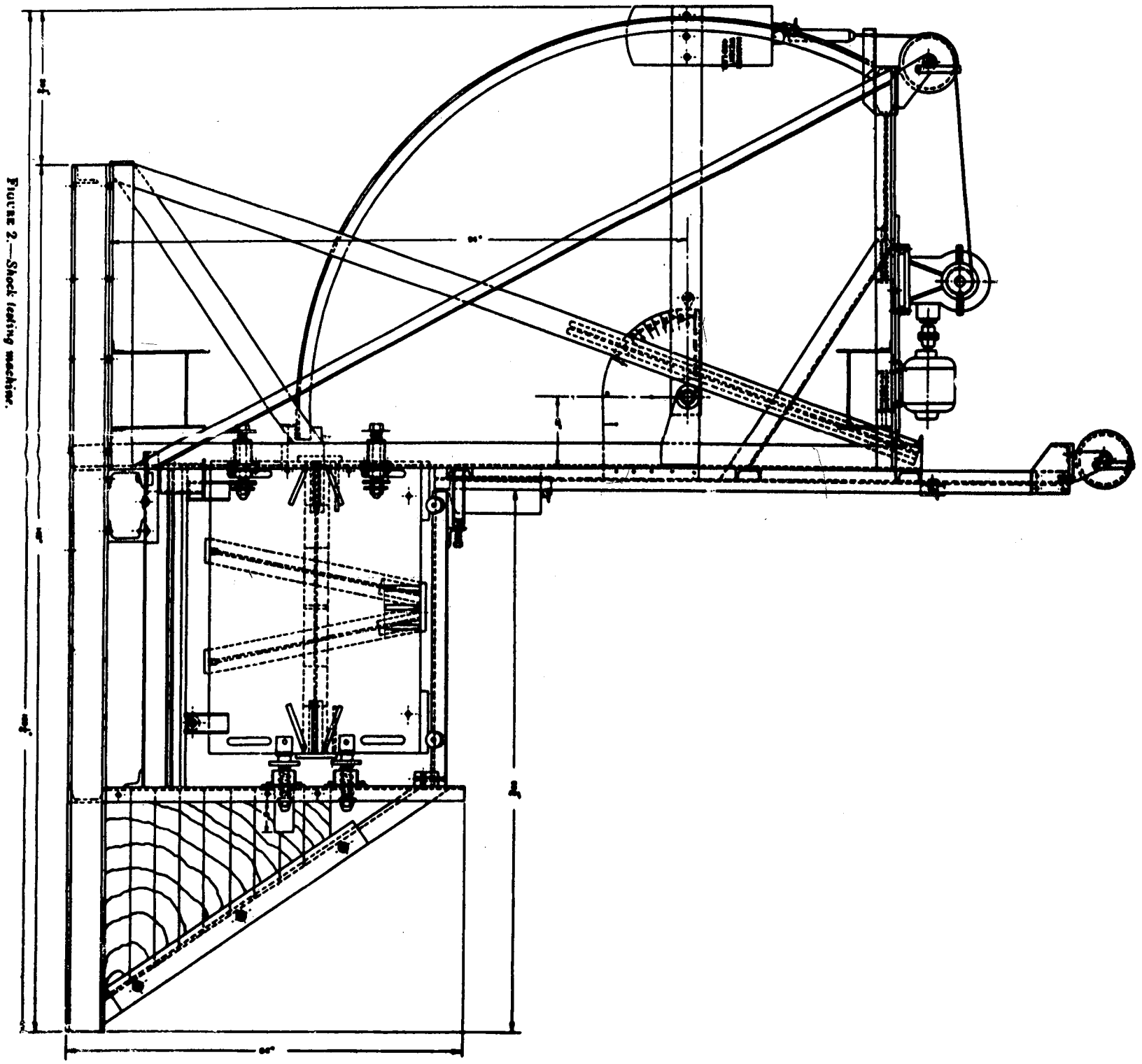


FIGURE 2.—*Shock testing machine.*

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INDEX

	Paragraph
Accessibility	3.41.2
Altitude	3.46
Altitude test	4.4.4
Atmospheric conditions	3.46
Batteries	3.52
Bolts	3.21
Cables, interconnecting	3.4
Cabling	3.34.6
Capacitors	3.6
Capacitors, mounting	3.6.5
Case	3.51
Ceramics	3.14
Circuit breakers	3.42.8
Coils, choke	3.13.3
Combination case	3.51.2
Composition of test set	3.49
Compounds, screw-thread retaining	3.16.3
Conductor insulation	3.33.2
Connectors, electrical	3.4.5
Connectors, interconnecting	3.4
Controls	3.7
Corrosion resistance	3.1.2
Cotton	3.14.10
Crystals	3.9
Definitions	3.2
Design of test set	3.41
Diagrams, schematic	3.39
Dielectric materials	3.14
Dimensions	3.54.2
Dynamotors	3.10
Enclosure	3.50
Fabrication	3.54.3
Fasteners	3.12
Felt	3.1.5.1.2
Finishes	3.35
Flux, soldering	3.25
Fungi test	4.4.3
Fungus conditions	3.46.2.4
Fungus-inert materials	3.1.5
Fungus resistant treatment	3.36
Fuses	3.42
Gears	3.54.6
Handbook stowage	3.40
Heater, standby	3.44
Humidity	3.46
Humidity test	4.4.2
Identification	3.38
Impregnating compounds	3.14.4
Inductors	3.13.4
Inspection tests	4.3
Instruction sheets	3.39
Instrument case	3.51.3
Instruments, electrical indicating	3.18
Insulating materials	3.14
Insulators	3.15
Insulators, shock and vibration	3.23
Interchangeability	3.41.4
Interference, radio	3.43
Jacks, audio frequency	3.4.5.3
Life, operating	3.41.5
Linen	3.14.10
Locking devices, thread	3.16
Lubrication	3.41.7
Magnesium	3.17
Marking	3.38
Material	3.1
Metals, dissimilar	3.1.3
Meters	3.18
Moisture pockets	3.41.6
Moisture resistant treatment	3.36
Mounting	3.53
Name plates	3.38.1
Nomenclature	3.37
Nuts	3.21
Nuts, castellated	3.16.2
Nuts, self-locking	3.16.2
Paint	3.35
Parts, miniature	3.3.5
Parts, nonstandard, approval of	3.3.3
Parts, nonstandard, replacement of	3.3.4
Parts, requirements	3.3
Parts, selection	3.3.2
Parts, standard, in re-orders	3.3.7
Plastic materials	3.14
Plating	3.35
Potting compounds	3.14.4
Power, input	3.48
Preproduction tests	4.2
Preconditioning for test	4.4.1
Protection of parts	3.42.4
Protection of personnel	3.42
Qualified products	3.3.10
Relays	3.19
Resistors	3.20
Rheostats	3.20, 4.5
Riveting	3.54.5
Safety	3.4.2
Screws	3.21
Screw assemblies	3.54.4
Screw threads	3.22
Screw threads in nonmetals	3.16.4
Serial numbers	3.38.1.4
Shafts, control	3.8
Shielding	3.43.2
Shock	3.47
Shock test	4.4.6
Snapslides	3.12
Sockets, tube	3.31
Solder	3.25
Soldering	3.24
Springs	3.26
Staking	3.16.3
Studs	3.12
Substitution of parts and materials	3.1.7
Switches	3.7.4
Switches, standard	3.27
Switch, power	3.45
Tape, electrical	3.11
Temperature	3.46
Temperature test	4.4.4

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	Paragraph
Terminals	3.28
Tests, classification of	4.1
Test points	3.42.2
Tools, special	3.29
Toxic materials	3.1.4
Transformers	3.13
Transformers, tapped	3.48.2.1.1
Transformers, variable voltage	3.48.2.1.1
Transit case	3.51.1
Tubes, electron	3.30
Variacs	3.48.2.1.1
Vibration	3.47
Vibration test	4.4.5
Washers, lock	3.16.1
Waveguides	3.5
Welds	3.32
Windows, instrument	3.18.2
Wires	3.33
Wiring	3.34
Wiring, safety	3.16.2
Wood	3.1.5.1.3
Workmanship	3.54
Zinc parts	3.1.6

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A. GIVE PARAGRAPH NUMBER AND WORDING.

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